

Futuristic view of the Internet of Quantum Drones: Review, challenges and research agenda

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ABSTRACT

The disruptive technology of unmanned aerial vehicles (UAVs), or drones, is a trend with increasing applications and practical relevance in the current and future society. Despite the common interest in drones for commercial deliveries, the use of this disruptive technology can be examined in the contexts of other world strategic demands such as climate change issues and traffic management. As of very recently, some drone-related futuristic disruptive technologies, including quantum drones (QD), the Internet of Quantum Drones (IoQDs), and a constellation of quantum satellites (CQS), are expected to be a breakthrough technology in strategic areas of society. However, prior research has not adequately examined and explored the potential applications of these drone-related futuristic disruptive technologies for social concern. Therefore, this study investigates how QD, IoQDs, and CQS can be applied in new contexts in real-time applications in strategic areas of societal interest, especially during the quantum age. Overall, our results unveil new potential and advanced applications to drone-related disruptive technologies in recognized and new contexts. Two relevant implications are highlighted. First, this research inaugurates new contexts regarding the use of drone-related technologies enabled by the Internet in themes of economic and social concerns. Second, from a futuristic point of view, the study examines the main challenges, risks, and advantages of the practical use of these technologies. We conclude this research with a summary of the main gaps and scientific challenges to the field and propose opportunities for future research.

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1. Introduction

While Paul Benioff introduced the principles of quantum information in 1980, it is possible to say that quantum computing developments started with Richard Feynman's works [96], fostering non-classical physical investigations [96,97]. Furthermore, Shor's integer factorization algorithm contributed to quantum computing showing the impacts of quantum mechanics applied to classical problems [97]. Qubits or quantum bits form a two-state quantum-mechanical system used by quantum computers. Qubits can be of various types, including charge, flux, phase, transmon, and others.

Qubits can be simultaneously zero and one, and a probability factor is always associated with the outcomes [98].

Admittedly, a number of computational works may be performed exponentially faster on a quantum processor than on a traditional processor [99]. Utilizing a processor configured with superconductors qubits, researchers [100] were successful in establishing quantum states on 53 qubits in a recent study (equivalent to a computational state-space of dimension (2^{53})). According to the results, the processor takes 200 seconds to sample one instance of a quantum circuit a million times, while a state-of-the-art supercomputer could spend 10,000 years to complete the same mission. Quantum computing has also enabled disruptive atomic-scale technologies [101] as well as natural and artificial atoms, as qubits accelerate the progress of quantum computation [102].

In recent times [98], experiments have been performed to transfer quantum encrypted messages for confidential information sharing. This experimentation is performed using multi-state en-

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Acronym	Description
CQS	Constellation of Quantum Satellites
DTN	Delay Tolerant Network
DiS-IoT	Direct-to-Satellite Internet of Things
ECC	Elliptic Curve Cryptography
HAQDN	High-Altitude Quantum Drone Network
IoD	Internet of Drones
IoP	Internet of Planets
IoPT	Internet of Planetary Things
IoQDs	Internet of Quantum Drones
IoQSS	Internet of Quantum Satellites
IoT	Internet of Things
LAQDN	Low Altitude Quantum Drone Network
LEO	Low Earth Orbit
LoRa	Long-Range
LPWAN	Low Power Wide Area Network
MiM	Man-in-the-Middle
NISQ	Noisy Intermediate-Scale Quantum
NB-IoT	Narrow Band-Internet of Things
NOMA	Nonorthogonal Multiple Access
QC	Quantum Computing
QC-MDPC	Quasi-Cyclic Moderate Density Parity Check
QCS	Quantum Computing Software
QD	Quantum Drones
QEC	Quantum Error Correction
QKD	Quantum Key Distribution
QPS	Quantum Positioning System
QW	Quantum Warfare
RFI-QKS	Reference-Frame-Independent Quantum Key Distribution
SA-MC	Simulated Annealing and Monte Carlo
TCP	Transmission Control Protocol
UAVs	Unmanned Aerial Vehicles
UDP	User Datagram Protocol

tangled qubits to ensure higher security standards compared to classical encryption approaches. In quantum networks, fibre optics and satellite communications are viable options for exchanging qubits. However, the present qubit exchange has various performance constraints expected to be improved in the future. Using drones, qubit exchange can be helpful in various applications like space surveillance and coastal water monitoring [103]. In these applications, deployment of multiple drones and constructing drone groups or swarms are necessary. In drone groups or swarms, trust-

based collaborative efforts are more useful. Here, trust is developed using the drone's functional behaviour analysis. For example, applying machine learning to analyze the drone's behaviour is studied in-depth [103]. In these applications, deployment of multiple drones and constructing drone groups or swarms are necessary. In drone groups or swarms, trust-based collaborative efforts are more useful. Here, trust is developed using the drone's functional behaviour analysis. For example, applying machine learning to analyze the drone's behaviour is studied in-depth [103]. Presently, drones are used as a medium for establishing quantum-based communication. Hence, any discussion over drones-supported by quantum technology means drones as a medium for communication. With the possibilities of this type of communication, it is expected that quantum device-integrated over drones for any storage, communication or computation will be developed in future.

Quantum computers or processors rely on three quantum mechanics in quantum information processing: discretion, superposition, and entanglement [99]. However, ethical issues emerge and should be considered despite the potential and advantages of these technologies in isolation or integrated with other disruptive technologies. In this regard, the scientific community has proposed that the term "quantum advantage" replace the term "quantum supremacy" [100] to prevent both false promises and misinterpretation of this technology [99]. The IoQDs is an example of a quantum drone-related futuristic disruptive technology that will be a breakthrough in strategic areas of society. This work has developed the IoQDs concept in multiple dimensions. Table 1 shows the comparative analysis of classical and quantum drones.

1.1. Internet of Quantum Drones

The application of drone-enabled IoT involves many security and authentication issues. In recent experiments [98,104,105], prototypes of drone-based quantum network communication have been discussed. In these prototypes and small-scale implementations, quantum signals are used over a small distance to exchange the qubits. With this feasibility, hovering the drones for quantum information exchange can easily be observed for any set of appli-

Table 1
Comparative analysis of classical vs. quantum drones.

Feature	Classical drones	Quantum drones
State	A bit in classical drone communication can either be 0 or 1.	A qubit can have multiple states, and it can be represented by 0 and 1 at the same time.
Power Consumption	It is directly proportional to the number of transistors in use. Thus, it gives a linear relation.	It increases exponentially with an increase in the number of qubit usage.
Error Rates	Classical drone-based communication has fewer error rate probabilities.	Quantum drones have higher error rate probabilities.
Operating Conditions	Classical drones can operate in normal conditions.	Quantum drones are required to have an ultracold environment.
Type of Computing	Logical operations are used in classical drones.	Unitary operations are used in quantum drones.
Gates Configuration	Gates in classical drones run in forwarding directions.	Gates in quantum drones are reversible in nature.
Bit Operations	It uses a truth table for bits-based operations.	A unitary matrix is used in qubits for bits-based operations.
Speed	Classical drone processors have comparatively lesser speed. For example, an n-bit processor can do one operation at a time.	Drone processors used for quantum communication require higher speeds. For example, an n-qubit processor can do 2n operations at a time.
Error Correction Possibilities	Classical computations are very advanced, and various approaches are available to correct the errors, if any.	Quantum computations are not advanced in handling error corrections. Thus, it is very difficult to correct the errors.
Security Primitives and Protocols	These are based on mathematical algorithms.	These are based on quantum properties.
Bits-Based Copy Operation	In the classical approach, copying bits is easy, and there are no restrictions.	Copying qubits can change the qubit states. Thus, this operation is not recommended.
Ultracold environment	The ultracold environment is dependent on the number of transistors present. With an increase in transistors, size, weight and power, consumption also increases.	Power consumption of quantum drones may increase exponentially for some real-time applications; thus, low size, weight and power drones are the need of the future. An alternative approach is to add an additional ultracold environment system to operate high computational or communicational quantum drones. For example, ultra-cold cesium atoms [169], ultra-cold atomic and molecular quantum gases [24], ultra-cold trapped ions [24], and ultracold atoms produced by evaporative cooling [200] are some of the sources that could provide ultra-cold environment to QDs.

cations. Since drones can be positioned or re-positioned anywhere, at any time, the IoD is very helpful for the city and rural areas. IoQDs provide the following primary features:

With a secure message exchange, the encrypted message can be safely transmitted via quantum communication between any two entities (devices or users). This message can preferably be exchanged through either satellite-based communication or fibre optics. With fibre optics, the loss of data is much higher compared to open environment communication using satellites. However, satellites are expensive. Further, satellites are less adaptable to on-ground dynamic situations.

With regard to the fast communication process enabled by IoDs, Lucamarini [106] identified that the optical-fibre-based quantum cryptography technique could work over 421 km. Additionally, the data transmission rate is much faster than existing long-distance demonstrations. Concerning the deployment to the city and rural areas, an intercontinental successful data exchange experiment was discussed using quantum mechanics [107]. This data exchange experimentation is performed for image exchange, and it was later extended for videoconferencing. Likewise, the experimentation can be repeated for short and long-distance communications to ensure city and rural network services. In a city network, fast and secure data exchange can help smart application development for traffic management, autonomous vehicle movements, transport network, healthcare services, etc.

Concerning the feasibility of developing and deploying private IoQDs, quantum drones provide full flexibility to constitute private IoQDs. These private IoQDs networks can be utilized in many useful real-time applications, such as military surveillance, traffic monitoring, Internet services, and secure long-distance message exchange.

1.2. Research motivation

This article is driven by the ability of quantum drones to do computations at a far higher level than conventional drones. The creation of scalable QDs-based computing technology will likely occur in a near future due to a variety of experimental initiatives and applications. Therefore, this study looks to be seeing an increase in the applications of secure and flexible drone movements and the constellation of drones with quantum communication support. It is conceivable to achieve scalable QDs and satellites-based computing if the QDs-based networks are capable of doing massive quantities of computations rapidly and effectively while keeping a low level of inaccuracy in the output for various applications like climate change, mobile networks, military purposes, internet services and many more. Accordingly, analyzing the scalability and security of QDs to integrate with IoQDs by wireless connections in the context of other needs is one way to approach the problem when analysing a demand to provide secure and faster communication to real-time applications. Therefore, the primary objective of this study is to provide a detailed discussion, comparative analysis, architectures and future directions on the subject of QDs, Quantum satellites and IoQDs for quantum computing and its futuristic needs.

1.3. Research contributions

The proposed survey covers: (i) the need for QDs, IoQDs and constellation of QDs from a futuristic point of view, (ii) the Design and feasibility of deploying LAQDN and HAQDN architectures for space communication. This type of communication is secure, faster, and covers very long distances compared to existing approaches, (iii) software life cycles, characteristics, applications and programming languages for quantum drones, (iv) The IoQDs and proposed

architectures and comparative literature analysis over these architectures are explored in-depth, (v) Challenges, issues, knowledge gaps and futuristic aspects of the IoQDs and related architectures are discussed in detail, (vi) Software characteristics, applications and Quantum programming languages are essential concepts that make the proposed survey different from existing work (for example, Kumar et al. [91]), (vii) IoQDs have applications in many important areas, including climate change, military, secure communications, aerospace and many more. This article discusses these applications in detail and (viii) IoQDs challenges and possible solutions for the future.

1.4. Organization of study

The rest of the paper is organized as follows. Section 2 presents the research methods applied. Section 3 conceptualizes and presents the importance of IoQDs and compares quantum drones with classical drones. Further, the importance of integrating quantum drones and satellites is explored. Section 4 shows the importance of software characteristics and aspects for satellite-based IoT environments or IoQDs, quantum software life cycle for quantum drones, and quantum automata. Section 5 presents the discussion section with futuristic scientific challenges in IoQDs. Section 6 presents the futuristic research directions and discussions. Finally, Section 7 concludes with the main findings of the study.

2. Materials and methods

2.1. Research method

This research examines how QD, IoQDs, and CQS can be applied in new contexts. This is examined for real-time applications in strategic areas of societal interest, particularly in the nascent quantum age. To address the research objective, a multi-stage research approach was proposed and theoretically examined. In the first stage of the research design, a systematic literature review was applied to identify, assess, and synthesize knowledge from the literature concerning the use of QD, IoQDs, and CQS. In this stage, the inclusion criteria for selecting the studies were based on keyword-based search and content analysis of the publications. In this case, the content analysis searched for studies on the application, usage and futuristic possibilities associated with “quantum drones”, “internet of quantum drones”, and “constellation of quantum satellites”. In the second stage, we selected essential information from the recent publications examined in these areas. This step allowed us to identify the primary challenges and opportunities for implementing these disruptive technologies. In this research stage, those articles were picked that addressed the needs of drones with computing abilities for real-time applications. Here, quantum computing-based abilities are given higher importance compared to other approaches. We employed an integrative approach review to synthesize the relevant literature in the field during the third search stage. The integrative approach mainly includes the possibilities of integrating two or more technologies with quantum computing, communication or storage power enhancement in future. Here, claims are proved with existing well-established theories. In the above three research stages, those articles are not considered in subsequent stages, which are not found to be presenting the futuristic directions of quantum technology-based approaches or applications. Next, the advantages and drawbacks involving the implementation of QD, IoQDs, and CQS were discussed, and we proposed a set of exciting research avenues for future research to address the most relevant challenges of scientific knowledge available. Strategies to overcome the main limitation associated with QD, IoQDs, and CQS were proposed.

2.2. Data collection and analysis

The primary data source was obtained through a systematic literature review [132]. Concerning the research objectives, we performed an integrative review to synthesize the accumulated state of knowledge on QD, IoQDs, and CQS. Integrative reviews establish discussions and summarize a determined theme from various publications in an integrated manner, leading to new perspectives. This approach is indicated for exploring novel topics in scientific literature. The search for relevant publications included databases, keywords, and eligibility/coding criteria. Complementarily, we also adopted the snowball process [133] to identify other relevant works and cross-references. To define the keywords and Boolean operators, a search process dedicated to finding the main terms associated with “quantum drones,” “Internet of Quantum Drones,” “quantum satellites,” and “constellation of quantum satellites.”

In the following step, the search process was carried out with the keywords on the databases Scopus and Web of Science. Next, the exclusion and inclusion criteria were applied to searching relevant publications. The eligibility criteria considered included relevant peer-reviewed papers published in journals and conferences testing or developing artifacts related to QD, IoQDs, and CQS; papers demonstrating experiments and results from the implementation of QD, IoQDs, and CQS; and literature reviews on the topics. Exclusion criteria removed purely theoretical studies, lacked methodological rigour or failed to provide contributions about the use of QD, IoQDs, and CQS in areas of economic and social concerns. The data obtained were examined in the following main phases. First, the relevant information from each article was selected. Next, a cross-analysis of the articles was conducted to find patterns of findings and information about the areas of economic and social concerns related to QD, IoQDs, and CQS. Third, a causal analysis among the set of information was conducted to look for a relationship between the findings. We then consolidated the findings and critically synthesized the main challenges and opportunities to QD, IoQDs, and CQS. The outcomes obtained are discussed in the following sections.

3. Internet of Quantum Drones and satellites

This section explores the application, opportunities, challenges, and future directions of IoQDs. Further, quantum satellites and their integration with IQDs are explored.

3.1. Internet of Quantum Drones: areas of applications and opportunities

In several studies [98,104,105], IoQDs were used to exchange confidential messages across the continents. These experimentations were conducted to transmit data over long-distance and with low-power technologies. These low-power technologies were based on the IoT with high precision in determining the drone's position. Some experiments with long-range, low-power technologies, based on the IoTs informing the position of drones with high precision, proved positive. The results indicated timely reporting of the position of drones in regions that WiFi and other technologies may not cover. These IoT technologies can be used, for example, in reconnaissance drones to locate other damaged drones in remote or unsafe regions [112]. Likewise, various real-time applications are feasible with the development of IoQDs networks. Fig. 1 shows different applications of IoQDs. Some of the applications where IoQDs will be highly useful are briefly discussed as follows.

3.1.1. IoQDs for radar and traffic management

Using quantum computers in quantum radar technologies aims to extend the precision in detecting and identifying the sensing

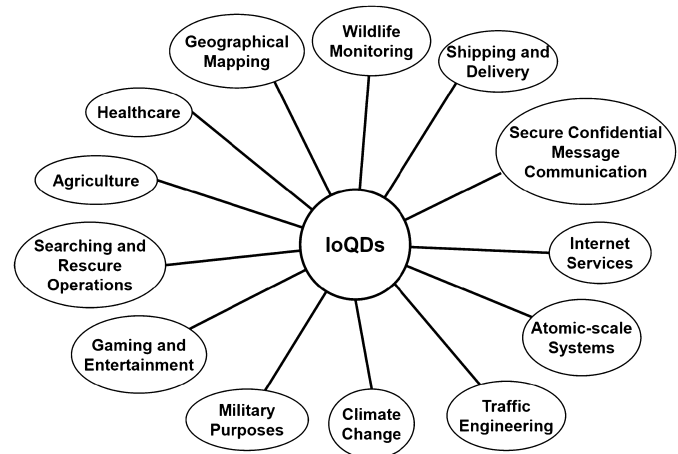


Fig. 1. Applications of IoQDs.

system (e.g., to pass/break through fog or clouds). Recent research looked at the benefits and drawbacks of quantum radar technologies [97]. The study discovered that quantum radars allow low-power operation consumption while maintaining optimum sensitivity and monitoring. However, future work must address inefficiencies related to obstruction, cancellation calculations, and factual examination of unused space. Typical areas of use of quantum radar technologies include military and civil questions. The results were also successful in an experiment on the universal representation of quantum radar cross-sections for 3D geometry complex targets (e.g., B2 aircraft). However, the quantum effect of sidelobes of bistatic quantum radar cross-sections in scattering is a common type of quantum interference and remains a challenge to the field [114]. Future research would be needed to advance practical applications by increasing the resolution beyond two-dimensional goals [113].

Fig. 2 shows the application of IoQDs networks in traffic engineering applications. Swarms of IoQDs will help exchange the on-ground vehicle data to long-distance traffic management-based data processing units. Traffic data can be analyzed in these units, and vehicle routes can be diverted to avoid congestion. The drone-to-drone and drone-to-ground station data exchange experimentation are found to be successful [98]. This experimentation can be extended to traffic engineering as well. Thus, it helps in efficiently handling the traffic, especially in congested cities, which, in turn, protects the environment as well. Fig. 2 is relevant for real-time handling of traffic or on-road vehicle movement scenarios in a secure and more efficient way. For example, faster qubit exchange between drones, drones and ground stations, or ground stations will add more accuracy in route establishment and secure data transfer. Furthermore, vehicle to vehicle communication through quantum drones will help in developing private vehicular networks, which will be helpful in various applications to handle privately owned vehicles. For example, ambulance services owned by a private hospital can be monitored through quantum communication-based private vehicular networks. In this network, message exchange will be more securely and improve QoS.

3.1.2. IoQDs for quantum Internet services

In one study [107], a satellite link was established for exchanging quantum encrypted data. This data includes images and video streams and is considered the first step toward “Quantum Internet (QI).” In this experimentation, an encrypted digital signals-based quantum mechanical process was applied. This process is applied to ensure the security of a large volume of data generated during Internet services. Digital signal-based encrypted quantum mechanical process makes the data exchange secure compared to the

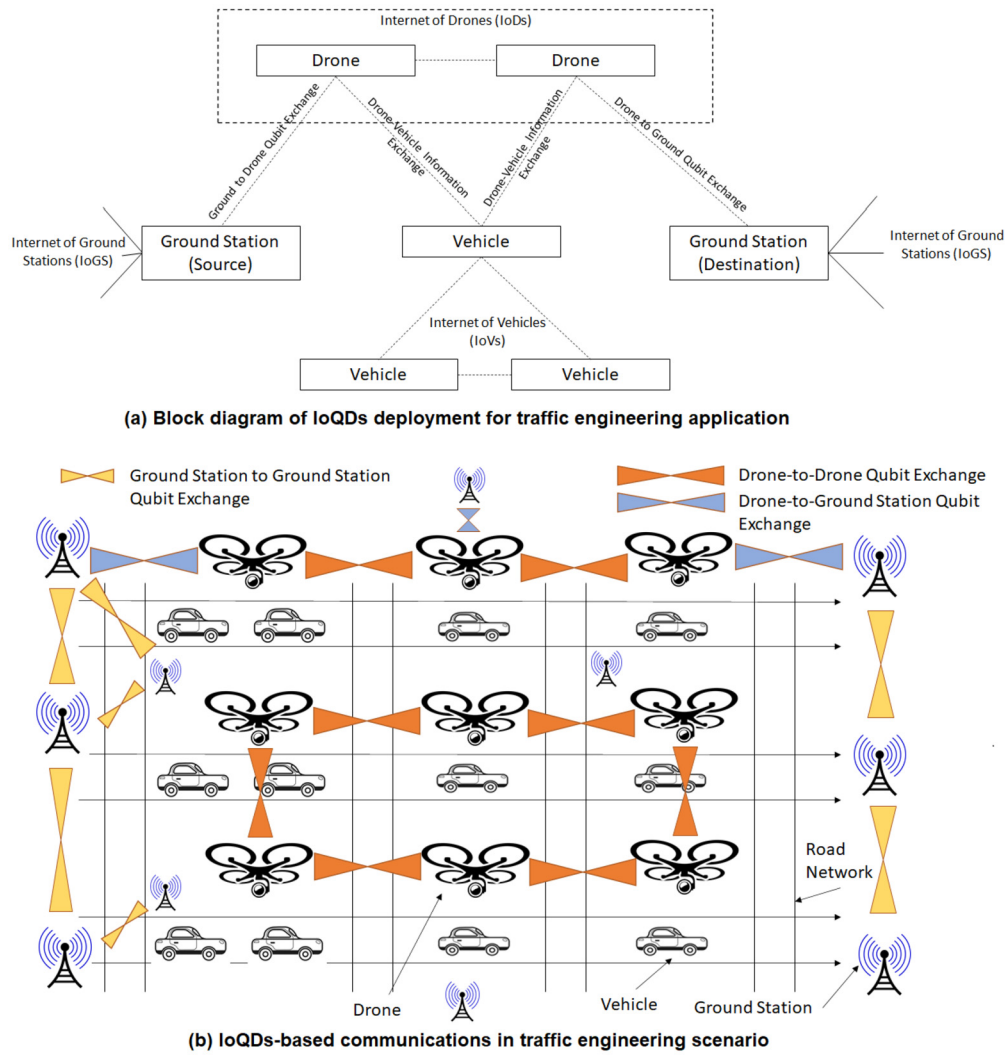


Fig. 2. IoQDs for traffic management.

classical signals-based approach. Quantum mechanics is a robust security process that makes the transmitted qubits difficult to intercept and read without being noticed at source or destination ends. Thus, it avoids attacks, especially MiM attacks or their variants. In intercontinental message exchange, China and Austria used satellite and fibre-optic networks for connection establishment, key exchange, and sending 5 kilobytes of image data. Later, this experimentation was extended to a video conferencing session (75 minutes in duration). The results of the observations were satisfactory for the very early stage of QI. Likewise, any private organization interested in establishing private and secure quantum-based internet services within their sister organizations placed in different continents or geographical regions can take the help of quantum drones to develop a network. Drone-based quantum internet services will be faster and more secure than traditional internet services. Further, the location of drones can be adjusted to get desired network connectivity. This will help to incorporate customers and stakeholders to give a provision in establishing business or interest-specific internet service groups with higher secrecy and better QoS availability.

3.1.3. IoQDs for climate change

Quantum computers and quantum computing are found to help handle climate-related data. In addition, data processing and fast computation make quantum computing an important candidate

for realizing climate change and associated incidents like hurricanes, tsunamis, earthquakes, and heat waves. If the information on these natural disasters can be processed in advance, many lives and much infrastructure can be protected. A constellation of IoQDs will help sense more reliable data and exchange it in multiple directions over a long distance. The feasibility of low- and high-altitude drone flying makes the availability of IoQDs-based climate data an essential aspect of disaster management. Fig. 3 explains the importance of drones in QC-supported areas. The essential systems and sub-systems were shown in Fig. 3, and their importance to IoQDs and climate change are briefly explained as follows.

IoT-based Network and Infrastructure: It is the consequence of both natural and human-induced processes that carbon dioxide emissions are produced. Decomposition, seawater release, and respiration are all examples of natural sources. Emissions created by humans include activities such as cement manufacturing, deforestation, power production, transportation, industrial sources, chemical manufacture, petroleum manufacturing, agricultural activities and the combustion of fossil fuels such as coal, petroleum, and natural gas. Gas that is harmful to the human body has a negative impact on health and adds to pollution. Public health specialists are concerned about the effects of pollution on human health. The production of coal-based commodities and feedstocks from carbon dioxide may be accomplished using renewable energy, which reduces greenhouse gas emissions while also meeting

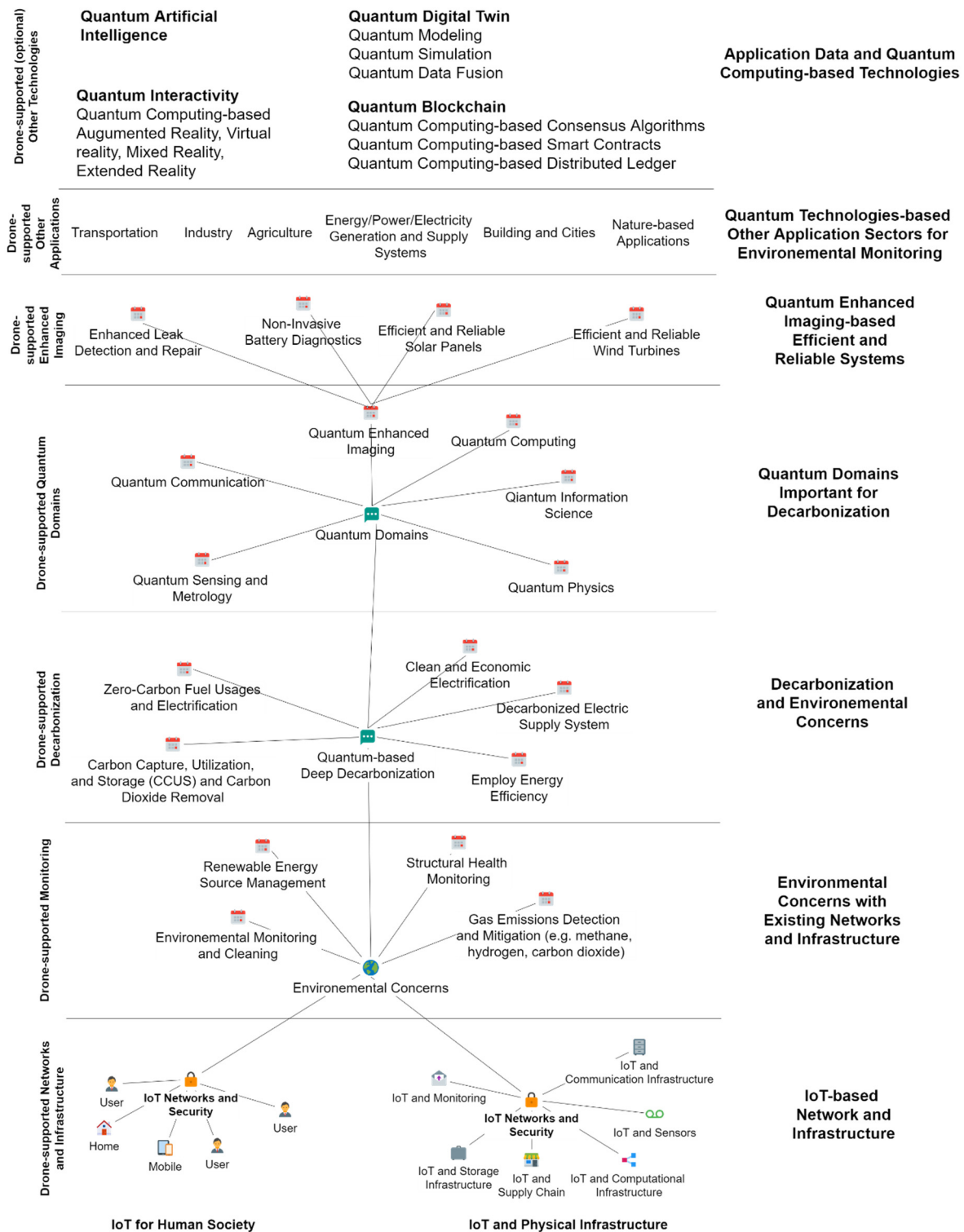


Fig. 3. Drones, QDs, IoQDs and Quantum Technologies for Climate Change and other Environmental Concerns.

the storage requirements of intermittent renewable energy sources. Here, Quantum-Dot-Derived Catalysts will be very useful for CO₂ Reduction Reaction [173]. For example, in the healthcare industry, quantum dots have been used in combination with other technologies such as the IoT and artificial intelligence to achieve astounding outcomes. With the aid of the information gathered and analysed, patient-centred solutions will be discovered, which will benefit the whole healthcare community. A Quantum-Dot system based on the Internet of Things will be referred to as a smart healthcare

provider, even if there are various restrictions, such as limitations on electronic waste and concerns about hacking [174]. Likewise, Quantum-Dot-Derived Catalysts will be useful in carbon dioxide and other harmful gases reduction in both human-based social applications and industry infrastructure. With the addition of drones, QDs and IoQDs, these catalysts will get additional support and mobility to cover a larger area in carbon dioxide reduction.

Environmental Concerns with Existing Networks and Infrastructure: Over the last several years, researchers have concentrated

their efforts on the unique thickness-dependent uses of two-dimensional materials, which include photocatalytic applications and materials that exhibit peculiar chemical and physical properties. A significant lot of interest has lately been generated in heterogeneous photo-catalysis, which has been piqued owing to the fact that it has the potential to address a wide range of energy and environmental challenges [175]. Likewise, heterogeneous photo-catalysts can be useful in other environmental concerns as well. For example, environments (water and air) that are free of pollution and clean energy are becoming more vital across the world, particularly in developing countries. Since semiconductor-based photocatalysis is both cost-effective and ecologically beneficial, a growing number of researchers are turning to this method of water pollution control in order to address the problem. Chemicals such as semiconductors-based photocatalysis are used to remove non-biodegradable inorganic pollutants such as pesticides and herbicides from water, but TiO_2 and ZnO have emerged as promising environmental remediation solutions due to their capacity to remove biodegradable organic contaminants from water [176–178]. QDs and loQDs can be useful in collecting water and air quality indexes from remote places, and QDs can be used with semiconductors-based photocatalysis for purification. There is a need to develop similar practices and experimentation in future.

Decarbonization and other Environmental Concerns: The transportation sector's rapid electrification and futuristic industrialization need quicker decarbonization. Improved energy efficiency is becoming more important to meet emission reduction objectives while simultaneously improving air quality and global temperatures. In the futuristic roadmap of various countries [179,180], Quantum technology is a viable option for decarbonization. With the support of QDs, loQDs, and quantum computing-supported technologies, environmental information can be exchanged securely and at a much faster rate. Further, decarbonization can also be performed at a much faster rate.

Quantum Domains for Decarbonization: The only option to significantly reduce greenhouse gas emissions is for us to change our energy consumption patterns and decarbonization. This action will have a significant positive impact on society and species. Thanks to quantum sensors, researchers may be able to get a deeper understanding of materials and quantum coherence processes. Nuclear reactors, for example, might be better monitored with the use of these processes and devices [181–185], which could also be used to search for new sources of renewable energy, such as geothermal energy. The use of quantum sensing to enhance maintenance and leak detection in fossil fuel infrastructure may assist in mitigating present system impacts as we transition away from fossil fuels in the future. However, the specific mechanism by which this would occur is unclear at this time. It is expected that various quantum domains will be associated with decarbonization in the future. For example, Berger et al. [181] associated quantum computation with climate change and decarbonization using chemical processes and materials. It has been discussed that gas-phase electronic structure, molecular dynamics, solution chemistry and vibrational and vibronic structure are important materials and chemistry simulations for carbon dioxide capture. Lah [186] discussed that the decarbonization of transportation in North America would be examined in great depth in future. The next stage will be for this group to examine the legal and institutional issues of the long-term sustainability of various climate change mitigation programmes. Following that, a proposal will be presented to the United Nations. It will be necessary to review previous studies to make sense of the findings and make judgments about which discoveries are significant for future studies. In these discoveries, quantum information science, quantum physics, quantum computation, quantum communication and many other quantum-related domains can play a significant role.

Quantum Enhanced Imaging and Associated Efficient and Reliable Systems: Quantum states of light imaging have the potential to outperform existing imaging technologies in terms of resolution, signal-to-noise ratio, and sensitivity, among other criteria. When it comes to quantum imaging, photon correlations are exploited to get beyond the fundamental limits of classical imaging [187,188]. These features give an advantage to quantum-enhanced imaging in the various systems to identify any leakages and protect turbines, solar and other systems [189,190].

Quantum Technologies-based other Application Sectors and Environmental Monitoring Concerns: Transportation, industry, agriculture, energy generation and transportation, infrastructure building, smart cities, and other nature-based applications are expected to use drones in the future at a large scale to improve all stakeholder experiences. Lau et al. [191] discussed that the transportation and power generation sectors are major sectors that are producing carbon dioxide at a large scale compared to other sectors. Further, it is expected that the percentage of carbon dioxide production will increase at a higher rate in future [192]. Thus, there is a need to identify the sources and decarbonize the environment. Here, a drone-based approach can help identify sources quickly from any remote location, and instruction or necessary steps can be taken to stop the leakage or production at the right time. From these research findings, Fig. 4 shows a possible scenario of how loQDs and the constellation of satellites can be formulated and integrated with ground networks for various applications.

Application Data and QC-supported Technologies: In advanced technologies, quantum blockchain [193–195], quantum artificial intelligence [196], quantum digital twin [197], quantum interactivity [198,199] and many more can play important role. For example, quantum blockchain for climate change observation would be useful in record keeping with transparency, immutability, security and distributed data processing and availability. This step will help in better record keeping and much faster and more secure data communication compared to traditional blockchain-based processes.

loQDs for military. loQDs enable a new field of application, namely QW. QW is a subfield in quantum technologies that employ Internet and quantum technologies for military purposes that affect intelligence, protection, and military and defence capabilities across all warfare contexts; the creation of new military strategies, doctrines, scenarios and attack capabilities; and peace or ethics issues. loQDs will improve the existing military intelligence, surveillance, target acquisition, and reconnaissance capabilities by providing more accurate navigation, ultra-secure communication, and computing capabilities [122]. loQDs also generate possibilities to constitute a private network for monitoring territories. This is an important application for any military network. In the present scenario, a large set of activities is executed with swarms of drones in military operations. With loQDs, faster and more secure data processing and transmission will be possible, which helps complete any important operation.

loQDs for mobile networks. Multi-state qubit feasibility and quantum computing features can enable fast data networks for mobiles. The existing cellular infrastructure can be used with quantum drones and quantum satellites. Additionally, security will be integrated to make a futuristic data network plan reliable at a much faster rate. An interesting work tested an optical quantum channel using wireless communication that can exchange quantum-secured random keys up to 10 kilometres by utilizing a network of several drones [115]. The system has shown that it can easily retrieve a signal that has been lost due to drone misalignment. Further developments include replicating the system in drone-to-drone signal locking and in-flight drone-based QKD. The feasibility of loQDs for mobile networks makes it possible to integrate the quantum-based internet service system with many real-time applications. For example, ad-hoc networks can

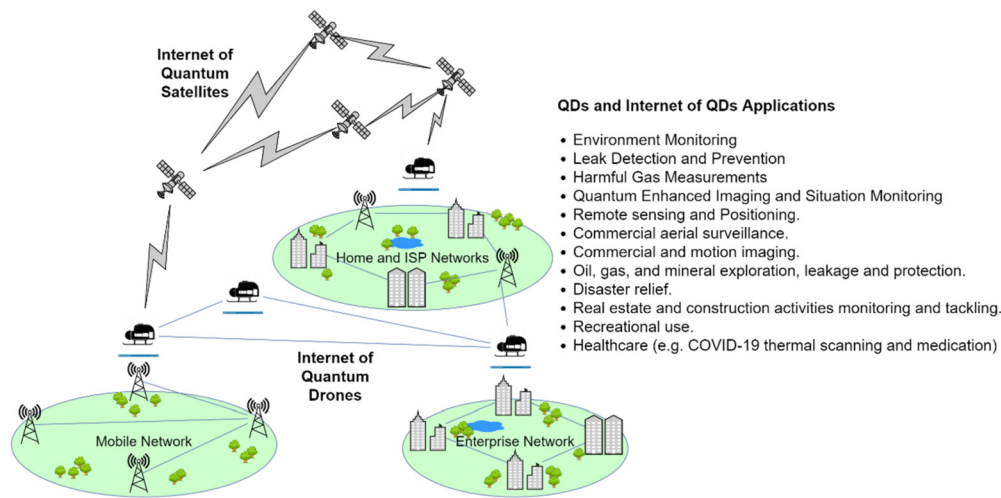


Fig. 4. QDs and IoQDs applications.

be constituted for vehicles to manage traffic and find routes using drones. Here, drones will be helpful in tracing any vehicles and ensuring emergency services with better location specifications. Furthermore, drones on-road can be helpful to identify law-breakers and follow the vehicles till the end. Isaac et al. [115] experimentation can be extended to offer private mobile networks using drones. These private mobile networks will facilitate business secrets, faster responses, independent control and flexibility to customize the network and its services as per dynamic requirements. Likewise, IoQDs or drone-based networks can be useful for other mobile network-based real-time applications.

IoQDs for atomic-scale systems. A new paradigm was recently tested for massive-scale atom handling by applying “molecular drones” [101]. The molecular drones create an individual atomic gap in a specific surface position, exploiting surface atom complexation as a fabrication tool. A breakthrough paradigm for big-scale surface atom manipulation using molecular drones was recently tested. Molecular drones use surface atom complexation as a fabrication technique to generate single atomic vacancies in specific locations on a surface. The experiments were conducted in an iodine-silver [110] altered coating as a substrate model. According to the authors, this disruptive technology could be used in other systems, e.g., non-metal surfaces. This new concept advances the limits of traditional scanning probe microscopies and serves as a foundation for developing potentially disrupting and entirely chemical, atomic-scale technologies [101]. As a result, quantum computers may be enhanced in the future by using hybrid technical devices that combine natural and artificial atoms and photons [102]. This new paradigm opens several breakthrough possibilities of applications of IoQDs in molecular drones in chemical processes and other atomic-scale systems.

Other applications. Among other applications, quantum drones can be used to ensure confidential message communication, shipping and delivery, wildlife monitoring, geographical mapping, healthcare, agriculture, search and rescue operations, and gaming and entertainment. Although traditional drones can be used in these applications as well, quantum drones will provide fast and secure operations (Fig. 4). Rahman et al. [166] discussed the blockchain integrated drone-based service system. Drones are becoming more popular for a wide range of commercial applications, including short-distance deliveries, and are becoming more affordable. In order to provide high-quality services to consumers and businesses, it is required to integrate a significant number of drone-based delivery service providers into the existing delivery service infrastructure, which is now under construction. This multi-drone movement needed authenticated drones to communi-

cate among themselves and ensure services. The use of blockchain technology has enabled the development of a drone flight enforcement tool, which guarantees that authenticated drone flights are conducted according to current norms and regulations [166]. Oubati et al. [167] explored that drones are already being used in many mission-critical applications. They may prove to be a more cost-efficient alternative to conventional aeroplanes in the future. For an example, being able to recognise an accident on the road, providing rescuers with their specific positions, and planning the shortest path to aid while keeping in mind the restrictions of the roadway itself are all important roadside assistance abilities to have. According to various professionals who have dealt with catastrophes in metropolitan areas, responding swiftly may significantly minimise damage and even save lives. Natural disasters such as earthquakes serve as the most convincing instances of this notion due to the dramatic nature of their events. These drones are capable of offering “bird’s eye view” viewpoints as well as adaptability, which allows them to be employed in a variety of different situations. In order to obtain information on interstate traffic flow, the researchers can use drones [168]. During their flight, drones will interact with vehicles already on the road to determine traffic flow on the routes they are travelling on. As a consequence of these modifications, the emergency response team will be able to react more quickly and efficiently. In addition, quantum drones can ensure that only authenticated drones follow pre-defined routes with more security and reliability. Quantum drones use a quantum key distribution mechanism for secure message exchange and authentication. Quantum cryptography or post-quantum cryptography mechanisms integrated with drone-based quantum communication are helpful in authenticated quantum drone network construction and secure message exchange.

3.2. Quantum satellites and optical fibres

Quantum satellites are a key component of a realistic free-space quantum network since they eliminate photon loss over long distances [108]. Quantum satellites and optical fibres are two prominent candidates for quantum signal transmission. In August 2016, China launched the world’s first quantum satellite (named Quantum Experiments at Space Scale (QUESS), nicknamed Micius) to create a secure communication system [109]. This satellite consisted of five major systems, including (i) communicator with quantum keys, (ii) emitter of quantum entanglement, (iii) entanglement source, (iv) processing unit, and (v) laser communicators, as shown in Fig. 5.

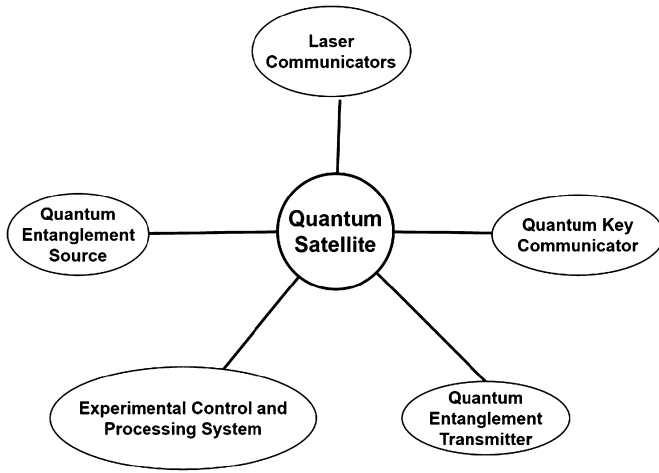


Fig. 5. Quantum satellite system [109].

Quantum key is theoretically considered secure because it is not possible to break it even with quantum computing approaches. The success of this experimentation and the feasibility of launching more such satellites will help create an loQDs network. This network will be helpful to establish secure communication for various applications in different domains like military networks, banks and finance, power grids, healthcare, Internet, telecommunication, and supply chain. A recent study [110] tested a method for a relay configuration of two quantum drones' signal repeaters located at different locations over the ground and one quantum CubeSat to solve the critical obstacles to the quantum connection between the Earth and a satellite (e.g., unfavourable weather and environmental conditions). The drones used in this experiment of quantum teleportation reached an altitude of 12 to 15 km and a maximum velocity of 482 km/h. The study suggests that the developed protocol allows spreading loQDs to all continents and nearby planets without being hampered by geographic features, environmental factors, or weather.

In another study [111], optical fibres were combined and linked to a satellite for achieving QKD over a long distance. Compared to satellite communication, optical fibres are less costly, provide high stability, and cover much longer distances for QKD. This experimentation has achieved a key generation rate 40 times higher than average. The fibre optical network and satellite integrated experimentation demonstrates that quantum interaction can be extended to large-scale real-time applications. The first drone-based entanglement delivery was recently demonstrated over 200 meters in all weather conditions with a 35 kg take-off weight [108]. The quantum network tested may be connected with image-drone dimensions for plug-and-play local-area coverage or installed in high-altitude drones for broad coverage, improving flexibility to the extant ground fibre-based quantum networks and satellites. The tests enable the development of a drone-based local-area quantum network with on-demand coverage of 40 minutes [108].

In Fig. 6, we demonstrated an example of an architecture for a long-distance connectivity system using drones, a constellation of quantum satellites, and an optical fibre network. With the success of the first quantum satellite and quantum drone [98,111], the feasibilities of installing multiple quantum satellites and the construction of a constellation of satellites increase to provide fast and long-distance services to many applications.

As shown in Fig. 6, a quantum satellite constellation can be constructed in space. Here, each satellite can communicate either directly with the ground station or through high- and low-orbit drones. Satellite communication can be used for key distribution, whereas fibre networks on-ground can be used for entangled photons.

3.3. Bridging knowledge gaps on Internet of Quantum Drones

Considering the importance of quantum drones and quantum satellites, this section considers the knowledge available about the architectures that can be used in the future for short and long-distance communications. loQDs-based short distance communication and its architecture will be useful for applications like traffic engineering, atomic-scale systems, internet services to single

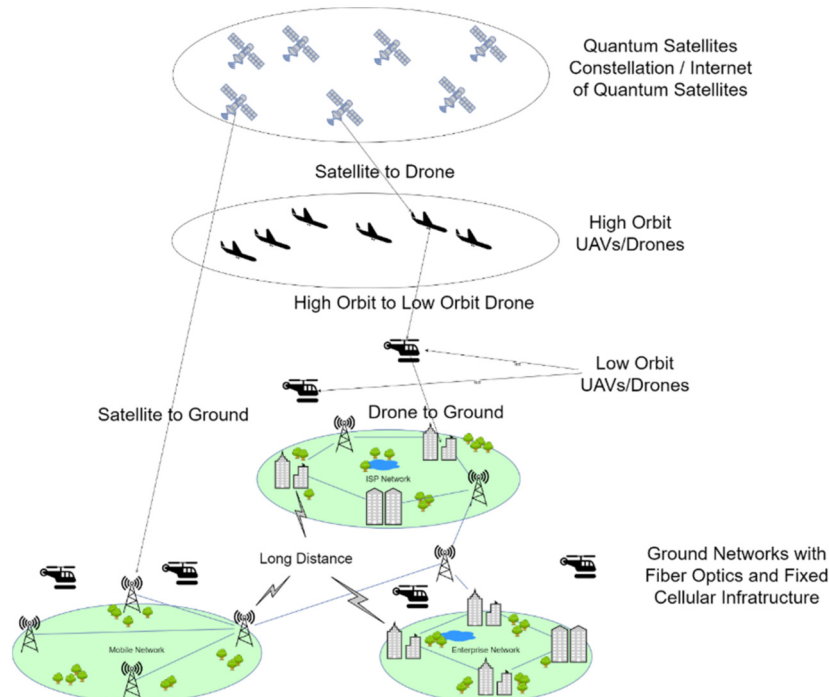


Fig. 6. Long-distance connectivity using quantum drones, optical fibre and satellites.

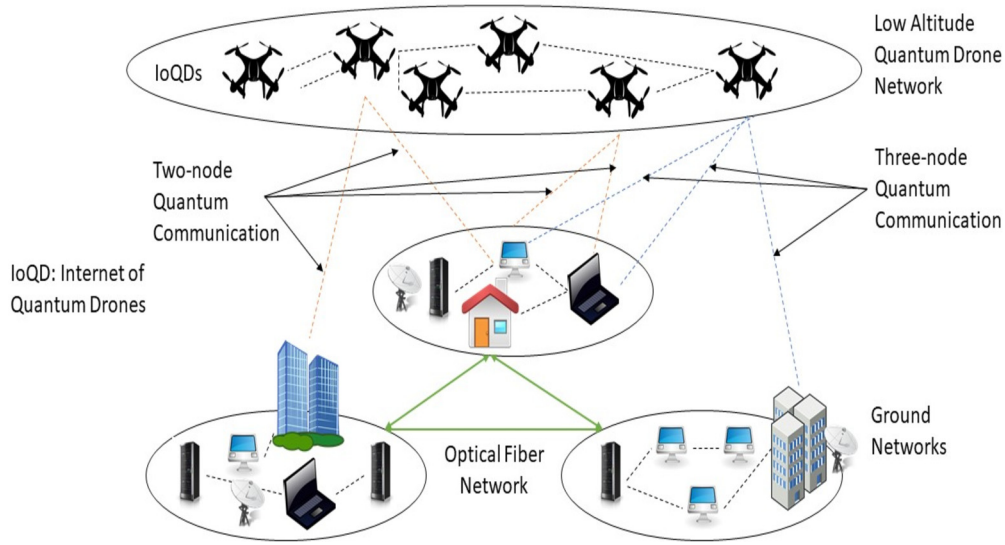


Fig. 7. Proposed LAQDN for short-distances communication.

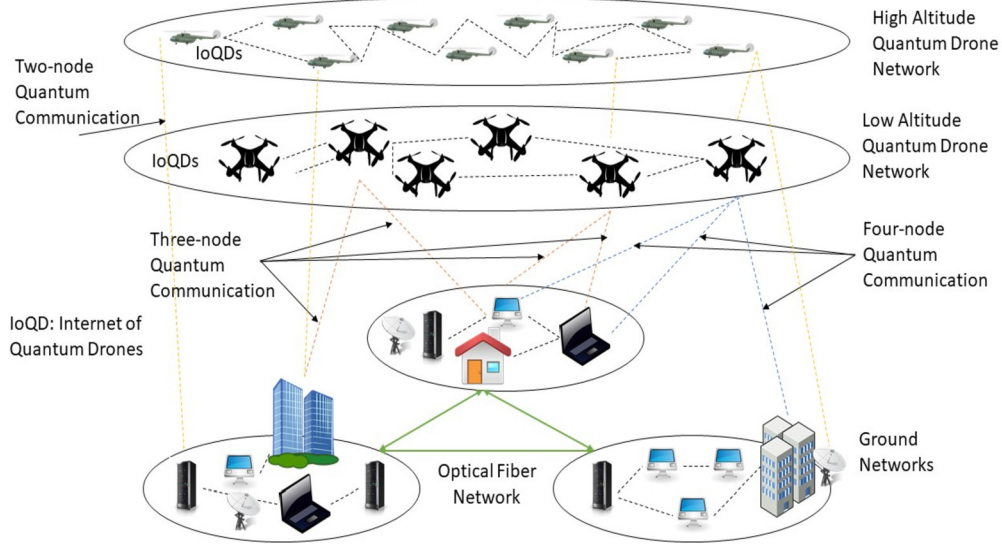


Fig. 8. Proposed HAQDN and LAQDN for medium-distance communication.

or group of buildings, private networks (e.g. shipping and delivery), healthcare, agriculture, searching and rescue operations, etc. Likewise, loQDs-based long-distance communication and its architecture will be useful for applications like secure key or message exchange between continents, shipping and delivery across different geographical regions, geographical mapping, secure military purposes, and many more. Different architectures are explained as follows.

3.3.1. LAQDN architecture

Recently, quantum drones have been used to enable short-distance ground-to-ground link establishment. Here, the quantum drone is considered an airborne node. Presently, this node is capable of maintaining two air-to-ground links for a very short distance (100 meters) [108]. Further, the quantum drone can operate in all weather situations. Fig. 7 shows the proposed LAQDN for short-distance communication within the local area, i.e. a few meters to kilometres.

Presently, experiments with two airborne nodes (quantum drones) within a few meters of distance have been successful. In the future, this can be extended to interconnect multiple airborne

nodes and formalize loQDs. loQDs will help in elongating secure data communication. To make this possible, however, there are many challenges. For example, airborne quantum signal storage devices must be designed and developed. Moreover, each airborne node should be equipped with temporary quantum signal storage capability. This is possible through quantum memories. Thus, quantum memories will be an indispensable component of airborne nodes to formalize long-distance loQDs networks. With quantum memories, quantum signals can be stored and retrieved as and when required. To ensure quantum memory over quantum drones, the major challenge is how to avoid the “entanglement-breaking channel” problem [130]. With the ability to operate a long time in the air and quantum memories, quantum drones will easily formalize loQDs. These loQDs will be able to serve on-ground real-time applications with less cost, more flexibility, high security, and fast communication.

3.3.2. HAQDN architecture

Fig. 8 shows the communication of LAQDN and HAQDN for short- to medium-distance (within a few tens of kilometres). In

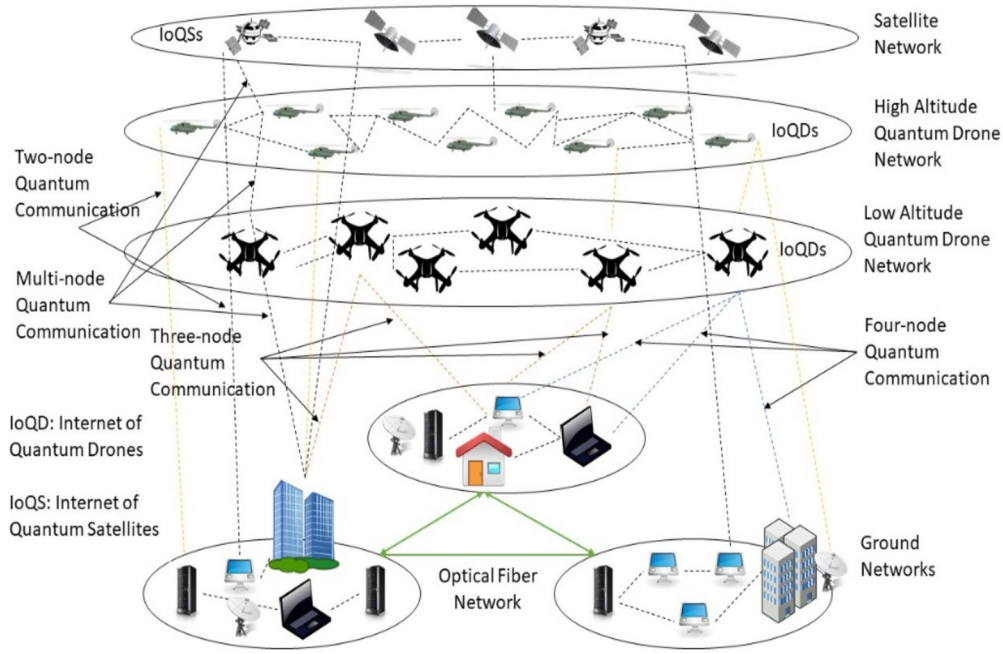


Fig. 9. Proposed satellite-based architecture for long-distance (within the country or between two continents) communication.

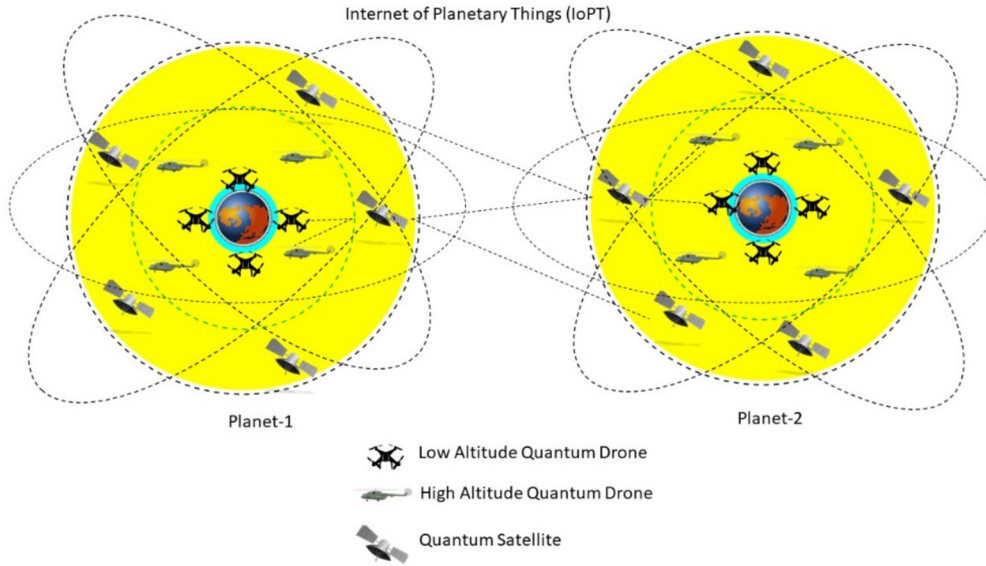


Fig. 10. Proposed planet-to-planet connectivity using HA.

both LAQDN and HAQDN, multi-node quantum communication is required.

Presently, a three-node quantum network has been found to be successful [131]. However, there is a need to extend this network for multi-node quantum networks. This is the base requirement to make the IoQDs network feasible at low and high altitudes. Additionally, drone at high altitude to drone at low altitude communication is required to be designed that make multi-hop communication feasible.

3.3.3. Satellite-based architecture

In satellite-based architecture, quantum satellites are also involved, as shown in Fig. 9. With quantum satellites, long-distance communication (inter-continental or inter-planetary) is feasible. This architecture is useful for collecting climate change information, conducting meteorological data analysis, predicting natural disasters, etc.

In this architecture, IoQDs and IoQSS will be required to make the proposed valuable architecture for real-time applications. In addition, this architecture can be extended to an IoPT, as shown in Fig. 10. In IoPTs, airborne nodes will help collect data from other planets or from space at large. Thus, a long-term space data collection procedure will be available without risking human lives. Planets may be able to speak with one another over the Internet in the future. The IoP is the term used to describe this phenomenon. It has moved to the top of the list of technologies for the IoT in recent years, and it is expected to remain there for some time [163]. The DTN is the most efficient technique when it comes to connecting things. There has been a significant amount of study conducted on IoP. Radio communication is the transmission of information using radio waves, and it is the fastest and most reliable kind of communication presently available. Because of the enormous distance between the two locations in interplanetary things becomes

one of the major drawbacks of radio-based arrangement, and it overshadows the other benefits. The current situation is that many research organisations and institutions are putting up significant effort to establish a feasible method of communication in space, which is currently considered impossible. The DTN communication protocol is especially well adapted for interplanetary communications, and it has been used successfully [163]. Other challenges to making IoPT possible are briefly discussed as follows [164,165].

In the past, astronauts from Goddard traded a file with the shuttle earth that travelled more than 600 kilometres above the earth's surface at the time. It was the first time a file from outer space made its way back to Earth without having its route set ahead of time [164]. To receive that transmission, technicians had to orchestrate things so that the link with the orbiting spacecraft was handed off like a cellphone transmission. Thus, covering the longer interplanetary distance is a major challenge. Alternatively, it is possible to expect severe transmission delays and other challenges involved with space communications, which have been previously reported. It is possible to deal with bad connections in space using two different approaches: UDP and TCP, and both of these methods are relatively straightforward to implement. However, there is a need to develop more such protocols that more efficiently deal with IoPT QoS parameters.

Another issue is the restricted supply of power, which is necessary for IoPT mission but also a contention source. When a spacecraft is running low on fuel, how can TCP or another protocol be made less communicative than it already is? This issue lies with other protocols as well. Thus, it needs to be addressed. Likewise, low power hardware resources for long-distance communication must be developed. An alternative to direct long-distance communication between source and destination is multi-hop topology, which uses low-power resources efficiently and for a longer duration. The lack of IoPT standards and semantics made this technically impossible. When it comes to space technology, the IoT may be able to seamlessly integrate with existing systems, thus opening the door to new possibilities for future space exploration and travel. The likelihood of a cyberattack is increasing directly to the availability and accessibility of IoT technology and applications in terrestrial locations. As a consequence, exploring the IoPT applications and protection from a cyberattack represent other significant challenges.

3.4. Cryptographic primitives and protocols for IoQD

A recent review study [116] examining the status of research on quantum-secure microgrids (i.e., microgrids that can withstand attacks from quantum computers) found several issues with the current quantum cryptography methods used in microgrids. While QKD can be considered the most mature technique in quantum cryptography, challenges to overcoming quantum-secure microgrids applying QKD are related to improving the resilience against side-channel, denial-of-service attacks, and more real-time experiments.

Another interesting piece of research [117] tested an earlier unconditionally safe ground vehicular to IoD communication framework. The study used IEEE 802.16d for long-range (up to 75 km) wireless drone-to-drone communication and security keys with digital signatures exchanges (based on the Kirchhoff-Law-Johnson-Noise (KLJN) scheme) between certification authorities for air-ground communications control. The two main advantages observed by KLJN were security and cost. The authors stated that hackers could not crack the key using an infinite computation resource such as quantum computers. Also, KLJN possesses a lower cost in comparison to QKD.

Lightweight cryptography approaches are preferred in drones to ensure energy-efficient, resource-optimized, cost-effective, and

comprehensive cryptographic primitives and protocols. In lightweight cryptography approaches, the major challenge is how to achieve performance with compact resource-constrained devices. Additionally, security is required for wireless communications between drones in flight. For example, several experiments are performed for secure QKD.

3.5. IoQDs assistance paradigm and quantum Internet

As discussed earlier, IoQDs are a new technology that can be used in a wide range of enterprises, human endeavours, space communication and inter planet information gathering. UAVs and drones are now being employed in a wide variety of sectors and human endeavours, including agriculture, healthcare, vehicular communication, disaster management and medicine delivery. Drones have been more popular as a fundamental concept in recent years, resulting in the development of novel links between terrestrial networks and floating UAVs in the sky. The feasibility of IoQDs makes communication more secure, faster and over long distances. Using such an approach has been more straightforward since wireless interfaces incorporated into communication devices have made it easier and more comfortable. Due to their variable heights and controlled movement, IoQDs may also be the most effective means of improving performance and overcoming the limitations of terrestrial networks in certain situations. The significant recent contributions to discussing the UAVs or IoQDs assistance paradigms are briefly discussed.

Alzahrani et al. [135] present the level of UAV-assisted research in fields such as routing and data collecting, mobile communications, IoT networks, and disaster management are sought to be analysed and characterised. This effort identifies and characterises the many ways present enabling technologies and especially UAVs to support these areas. This study represents a significant step forward because it is the first to conduct a comprehensive analysis of the many UAV-assisted networks currently in use. This work has explored the multi-layered approach for potential application cases. This work has explored information gathering, intelligence collection, network construction, routing feasibilities, data gathering, important technical and research finding for various on-ground applications. Gyongyosi [136] explored that entangled quantum networks are required for the operation of any quantum Internet on a global scale for it to be functional. In this work, a mathematical model is developed that can be used to analyse the dynamics of entangled network topologies and entanglement flow in the quantum Internet. To support quantum internet, QDs and IoQDs can play an important role. Networks using QDs and IoQDs can provide closer, private and secure quantum internet connectivity. As well as identifying and characterising the dynamics of entangled quantum networks, the model's analytical solutions can also identify and characterise the stability, fluctuation characteristics, and dynamics of entangled entanglement flow in entangled network topologies. A variety of entangled structures may be used to demonstrate the model's results and quantify the system's behaviour under consideration.

Gyongyosi and Imre [137] discussed that entangled quantum networks, which are the fundamental building blocks of the quantum Internet, are constructed using quantum entanglement as a fundamental building block. When it comes to quantum networks, the distribution of entanglements is a key and rapidly changing subject that is growing more important as time goes on. The study's approach for high-fidelity entanglement distribution is a basic solution that is easy to implement and manufacture. Therefore, an opportunistic-based approach was proposed for enhancing quantum internet services. In another research, Gyongyosi and Imre [138] discussed the quantum Internet's entangled network architecture, defined by a high degree of complexity, making getting

around it difficult. This is due to the quantum Internet's entangled network design and complexity. When it comes to the quantum internet, scalable routing is a routing approach that can identify the most optimal subnetwork routing in order to conduct a high-performance and low-complexity routing in the entangled structure of the quantum internet. As one of the routing algorithms suitable for usage in the quantum Internet, scalable routing may be used to identify the optimal path for a particular subnetwork. Additionally, it was proposed that, while working with a quantum network, an exploration approach for routing space be used in conjunction with a scaling mechanism for routing in the quantum network. Examples presented by Gyongyosi and Imre [138] include the fact that scalable routing may be used to provide compact and efficient routing while maintaining excellent performance in quantum networks.

Gyongyosi and Imre [139] discussed that quantum repeater networks would be critical infrastructure for long-distance communications and the future quantum Internet, and they will be required to be in place to function properly. The study indicated that depending on the degree of entanglement that occurs between the quantum nodes, a variety of network topologies, including multi-level networks, may be produced in the presence of entangled quantum nodes. They analysed a methodology that uses efficient routing algorithms to determine the shortest pathways in entangled quantum networks has been shown to be practical. Authors explored that base-graphs may incorporate entangled network topologies, which retain the probability distribution of connections while providing efficient decentralised routing and other types of networks. The study suggested that they might be used immediately for practical quantum communications, experimental long-distance quantum key distribution and repeater networks, as well as quantum repeater networks in the future, among other applications. Gyongyosi et al. [143] discussed that it might be possible to leverage the quantum nature of the information to increase data processing efficiency. Using quantum information processing, substantial improvements in computer science may be possible shortly. Quantum channels may transfer both conventional and quantum information, depending on how they are constructed. To better understand how quantum channels function, this work goes into great depth about how they operate and how different capacity measurements are carried out to understand better how they work. Quantum entanglement, also known as "superposition of states", means transmitting quantum information that cannot be traditionally expressed by merging two or more states. In contrast to computer networks, optical fibre networks or wireless optical channels may be used in the real world to build quantum channels instead of computer networks.

Wehner et al. [144] discussed that the internet's global network of immediate communication has radically changed our way of life. Even if two places are thousands of miles apart, quantum internet technology can help them all. Many tasks now out of reach with current technology may be possible in the future with a quantum internet. In this work, an outline of the procedures needed to implement a fully working quantum internet is discussed. Achievements in theory and practice are also explored. According to this work, quantum hardware suggested for quantum communication can be mapped to drone-based quantum communication. Fig. 11 shows Quantum drone and Quantum satellite-based scenarios and hardware requirements. As shown in Fig. 11, quantum satellite and quantum drone-based switch, repeater, channel and end node applications are required to construct quantum internet. This integration is possible with IQDs or IQDs. Also, a quantum satellite-based switch helps in multiple interconnecting satellites or quantum drones. The quantum satellite and quantum drone-based network required a quantum drone-based switch and repeater to expand the network and pro-

vide services to end systems. Pirandola et al. [145] found that the proposed investigations can correctly define quantum repeaters and set the limits of point-to-point quantum communications for repeaters. Already Pirandola et al. [146] explored that the adaptive protocols (teleportation stretching) may be decreased by using quantum channels, and inverse limitations can be imposed for quantum and private communications using quantum channels. Bosonic lossy channels are also subjected to extensive testing to verify that they have strong converse properties.

Other important quantum area advancements are discussed in the extant literature [100,147–151]. For example, Arute et al. [100] discussed the importance of quantum supremacy in programmable superconducting processors. Here, device design and architectures are explored in detail. Further, the importance of quantum circuits and single/multi-qubit implementation and error handling approaches are discussed in detail. Preskill [147] mentioned that NISQ technology would be accessible to the general public shortly. Here, it should recognise that NISQ devices represent a significant step forward in developing more powerful quantum technologies, regardless of whether or not they are ultimately usable. Harrow et al. [148] discussed that the goal of quantum supremacy is not to resolve a particular problem; it is instead to illustrate what conventional computers are incapable of performing under specified circumstances. According to the computational complexity concept, specific jobs are more challenging to do than others, and this may be shown rationally. The employment of complexity-theoretic assumptions that may be justified heuristically to achieve success in maintaining quantum supremacy will be required to succeed. In [149], quantum supremacy is explored in detail. Based on the most recent proposal from Google's Quantum Artificial Intelligence group, Aaronson and Chen [149] studied the issues associated with sampling the output distribution of a random quantum circuit. Here, it was observed that even though it has nothing to do with sampling, the natural average-case hardness assumption states that no classical algorithm can pass the same statistical test that the outputs of a quantum sampling approach pass. Neither the sampling method nor this premise is supported by the data available.

Additionally, Aaronson and Chen [149] proposed a new algorithm that simulates general quantum circuits with improved complexity. In order to advance quantum computing research, Alexeev et al. [150] conducted a thorough examination of scientific and community requirements, prospects, and obstacles during the next two to ten years. Gyongyosi and Imre [151] proposed a model of scalable distributed gate-model computation. It has been proved that the proposed architecture has an objective function that can be efficient for a computational problem if it lies in a distributed connected manner. Besides, the impact of decoherence was evaluated on this distributed objective function. Likewise, many advancements are made in quantum computer, circuit and communication areas. Among other approaches, Pirandola and Braunstein [140] discussed the possibilities of quantum internet and the importance of quantum teleportation in it. Lastly, the futuristic principles and options of quantum teleportation also are explored. Loyd et al. [141] discussed the need for quantum infrastructure for a quantum internet. Here, it is concerned that a team of academics from MIT and Northwestern University is investigating the possibility of teleporting quantum bits across long distances with high accuracy. It will be necessary for quantum computers to use a system like this to create their quantum communication internet. In this work, the most recent results and suggestions from the researcher's team are discussed in the teleportation architecture and its loss-limited performance analysis. Gyongyosi and Imre [142] surveyed the most recent advancements in quantum computing technology, and this study also assesses the remaining challenges in the area of quantum computing. The cutting-edge development in computer networking is quantum information processing. Over-

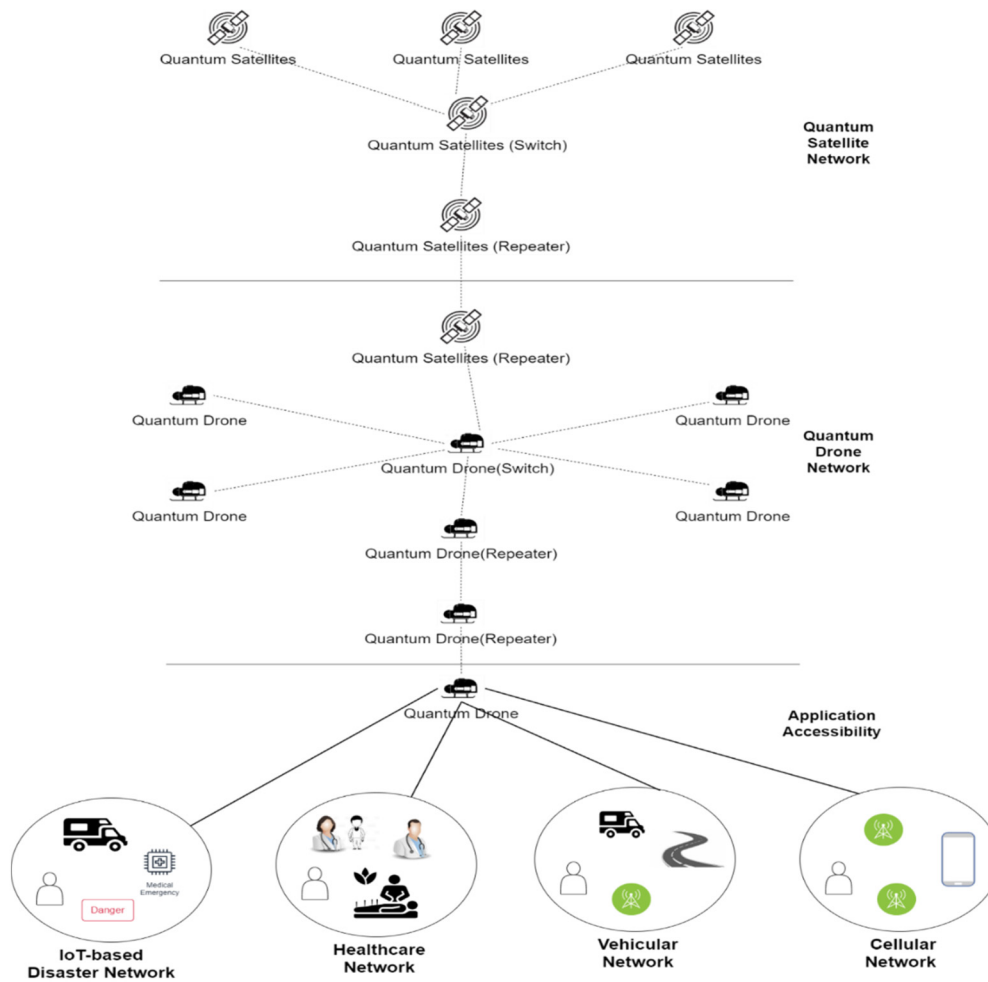


Fig. 11. Essential drone and satellite-based quantum hardware elements and proposed connectivity for ground-based applications.

all, the landscape of literature shows that there are several discussions about the future of quantum computing. Evidence from the literature suggests that computers may be reprogrammed via quantum computing.

In quantum entanglement, physical phenomena occur when a large number of particles come together in a way that each particle's quantum state cannot be described in isolation from the rest of the group. Fig. 12 shows a proposal of a multi-layered quantum entanglement-based approach for secure and faster communication. In this drone-based single or multi-quantum entanglement approach, quantum internet can be constructed, and services can be ensured for various on-ground applications. In this architecture, a quantum drone-based distribution point at the middle layer can integrate multiple-drone subnets and provide secure and fast quantum communication. In addition, the quantum internet layer may help identify the software and its life cycles necessary for internet services.

In quantum entanglement, physical phenomena occur when a large number of particles come together in a way that each particle's quantum state cannot be described in isolation from the rest of the group. Fig. 12 shows a multi-layered quantum entanglement-based approach for secure and faster communication. In this drone-based single or multi-quantum entanglement approach, quantum internet can be constructed, and services can be ensured for various on-ground applications. In this architecture, a quantum drone-based distribution point at the middle layer can integrate multiple-drone subnets and provide secure and fast quantum communication. The Quantum internet layer may help

identify the software and its life cycles necessary for internet services.

Likewise, in Fig. 13, we propose a multi-layered quantum entanglement-based approach using quantum satellites and quantum drones. In this approach, quantum satellite-based single and multi-hop quantum entanglement approaches can be applied to construct satellite distribution points. This makes quantum internet services possible through quantum drones and quantum satellites.

In this approach, multi-hop quantum entanglement satellites or drones are connected with other satellites or drones to ensure connectivity and communication. This connectivity and communication is a contact-based approach. For example, multi-hop satellite to multi or single-hop drone contact-based communication can constitute satellite to drone-based distribution point contact. Likewise, multi-hop drone to multi or single-hop drone contact-based communication can constitute drone to drone-based distribution point contact. Quantum satellites or quantum drones interconnecting two distribution points act as quantum repeaters. Contact-based temporal communication is an opportunistic quantum entanglement approach to constitute a more considerable quantum internet.

4. Software characteristics for satellite-based IoT and futuristic quantum-integrated benefits

This section presents a detailed analysis of recent studies on satellite-based IoT environments. In future, quantum satellites and/or quantum drone-integrated communications are expected to

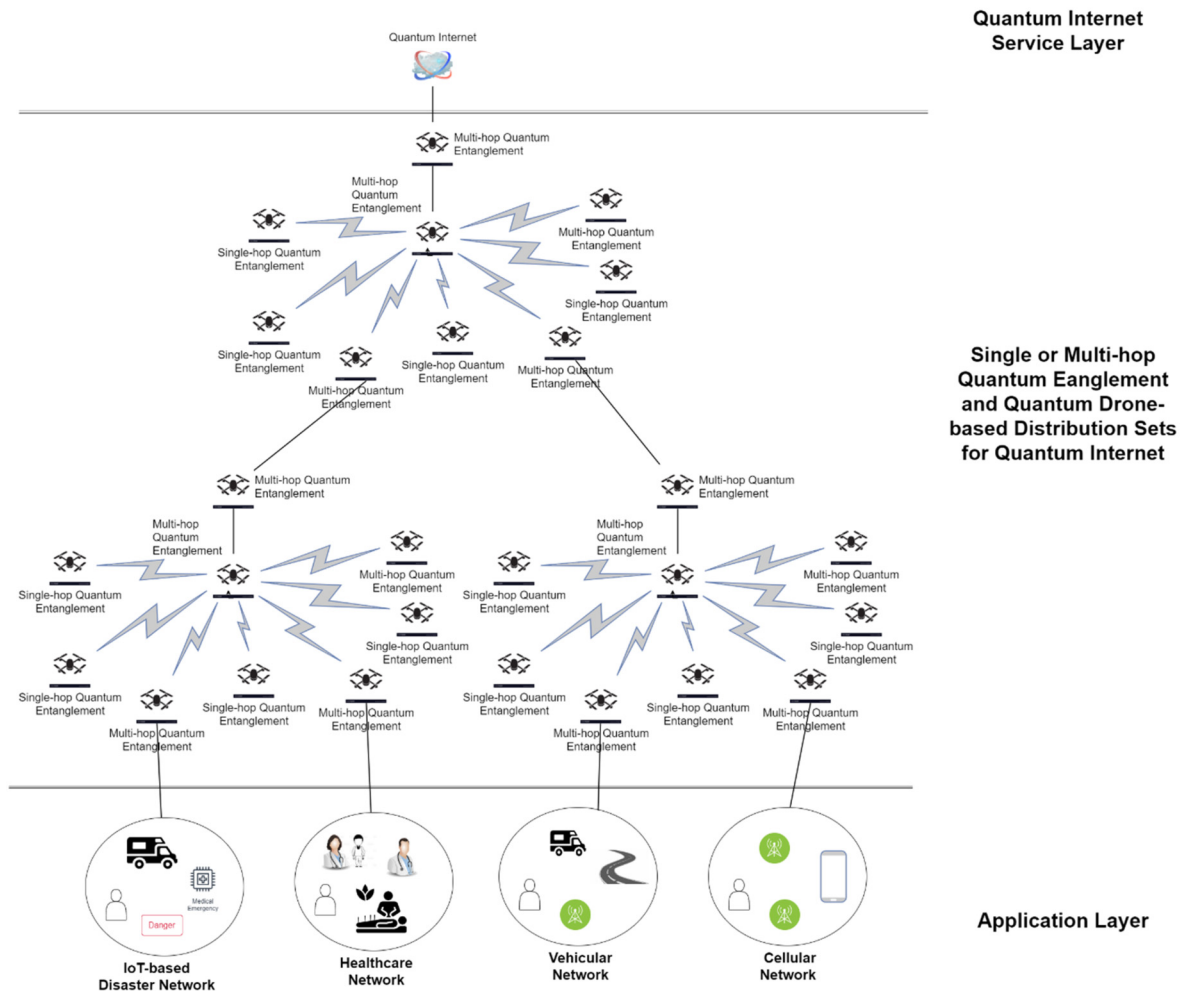


Fig. 12. Single and Multi-hop drone-based quantum entanglement for quantum internet services in different applications.

be explored in detail. However, a few initiatives are only taken to use quantum satellites to make it feasible [152–155]. Thus this section explores the existing satellite-based IoT network designs which can be transformed into quantum satellite integrated solutions in future. Here, a comparative analysis of those studies which perform quantum or software services to satellite-based IoT or constellation of satellites is performed.

Acín et al. [24] surveyed important quantum domains, including quantum communication, quantum computation, quantum simulation, quantum metrology, sensing and imaging, quantum control, and quantum software and theory. For example, superconducting circuits, electronic semiconductor qubits, impurity spins, and linear optics are considered essential fields in the quantum computation domain from a futuristic point of view. It has been concluded that functional quantum interfaces for short-, medium- and long-distance communication must be explored in detail. Moreover, algorithmic problems associated with the quantum computational domain are important for improving software and theoretical experience improvements. Likewise, other domains are explored as well. This work needs to be extended with a comparative analysis of different quantum domain features. In this case, characteristics and complexity-based analysis can be performed to analyze those technical aspects that need special attention for futuristic applications. For example, complexity and performance aspects in satellite-based IoT or networking aspects are found to be important to improve service experiences. Further, it has been revealed that certain satellite electronics have been rendered unworkable as a result of the radiation radiated from orbit [152]. When em-

ployed in the same manner as conventional electronic systems, nanoelectronic systems have the potential to improve the performance of electronic satellite systems. Here, QCAs (Quantum Cellular Automata) in nanotechnology can improve the performance. Thus, this research area could be more explored.

Chu et al. [1] discussed how fifth-generation (or beyond) multi-beam LEO IoT satellite communications systems are susceptible to a significant problem of channel phase uncertainty for beamforming, which is a recommended system for use with a LEO satellite. This work describes a multibeam LEO satellite network-based IoT network with a specific purpose. The article proposed two resilient beamforming methods to lower the overall power consumption of the LEO satellite when the operational channel phase is not favourable. The first option can be better utilized out of the two proposed approaches. Second-best options are always the best for crucial IoTs applications because of their functionality. Further, the design of robust beamforming for multibeam LEO satellite IoT network is formally designed and verified. The proposed study is simulated and experimented by analyzing the power consumption aspects. This particular study can be extended to include quantum computing and technology aspects. The literature also shows that the IoT network performance parameters can be considered to analyze the network's performance as a constellation of satellites. In [153], LEO satellites are explored with quantum communication. A significant role will be played by LEO satellites in the development of the global quantum Internet, which will be deployed in the not too distant future [153]. Because of the improved downlink, it is possible that methods that take advantage of this improve-

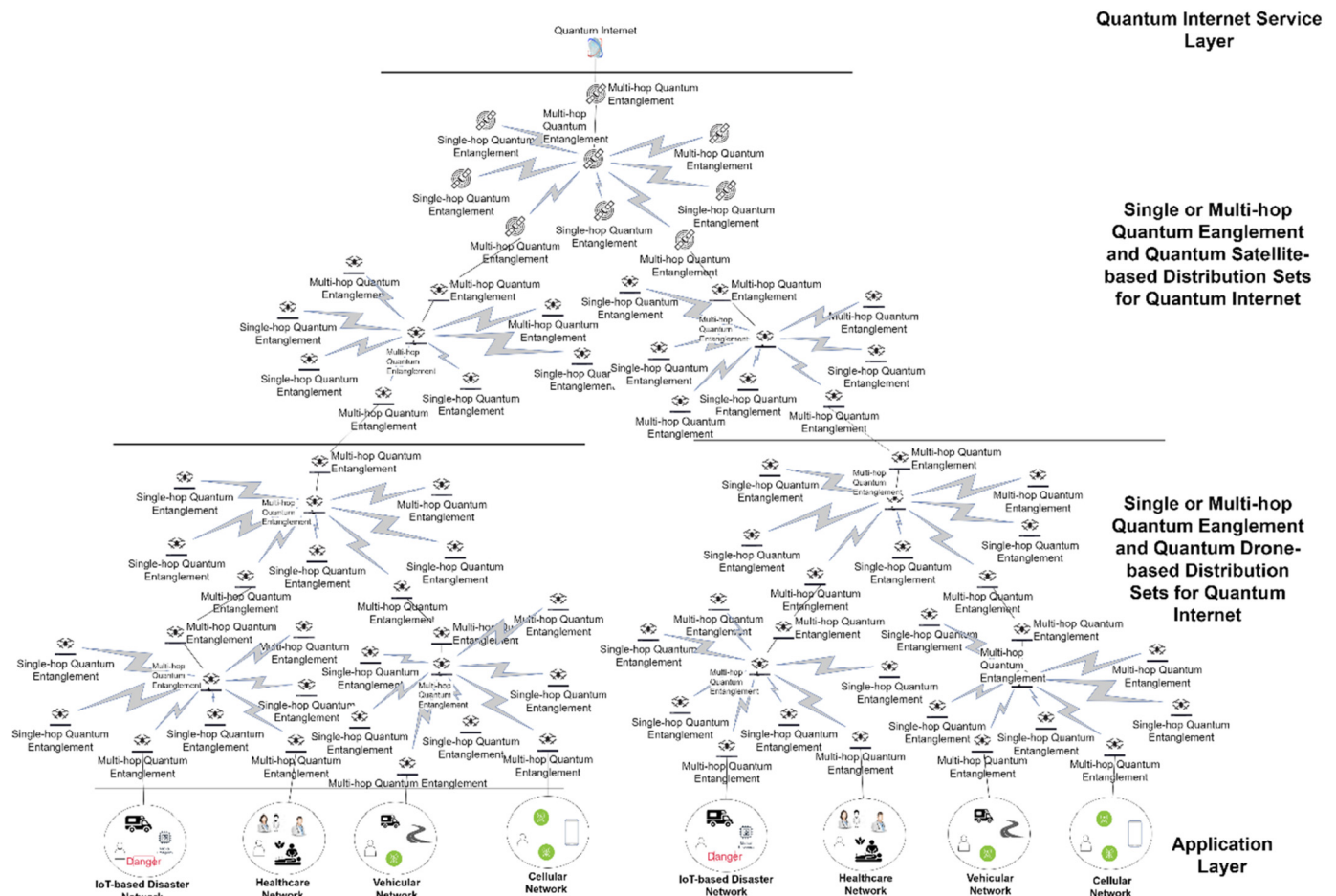


Fig. 13. Single and multi-hop satellite and drone-based quantum entanglement for quantum internet services in different applications.

ment will be able to enhance uplink quantum communication as a consequence. Designing future quantum Internet satellites should take into account uplink communication through quantum teleportation, which is currently being investigated.

Yang et al. [2] examined the architecture of the 5G mobile communication satellite infrastructure, including the Iridium, Orbcomm, NB-IoT, and Long Range Radio satellites. It is observed that NB-IoT system characteristics include optimized network transmission and low-power terminal design. These elements are used to construct a new dynamic routing protocol. The algorithm employed in the sensor network in this work is used in a wireless sensor network. In addition, this work has studied and investigated the effects of energy consumption and the number of node variations that remain in the wireless sensor network. Cryptographic techniques, such as ECC, are used to ensure the security of communication on the IoT and 5G mobile communication satellite infrastructure [155–157]. Despite the fact that public-key techniques like ECC are challenging to break, policymakers, scientists, and governments must begin preparing the Internet of Things for the quantum future as soon as possible. The specific objectives of this study are to investigate the impact of quantum computers on the security of cryptographic algorithms that are already in use and to present an overview of ideas for processes that are secure in both traditional and quantum computing contexts, among other aspects.

Graydon et al. [3] discussed the large population living in impoverished countries who do not have access to the Internet. Until recently, satellite Internet was the sole option for connecting homes and companies to the Internet, and it was only available

in specific locations. A number of organizations have claimed to be capable of providing low-cost, high-speed Internet connections to everyone. This work has conducted a thorough examination of the history of satellite Internet services and the current announcements of new satellite Internet service providers. According to the findings of this research, underprivileged regions would lose out on the advantages of satellite Internet technology if the government and corporate sectors do not follow through on their commitments. In discussions, it has been observed that the likelihood of satellite ISPs offering competitive services has increased in the last decade. However, there is debate about whether or not these new constellations are both technically feasible and cost-effective. To cope with the newly crowded orbital plane that will be inundated with thousands of additional satellites as a consequence of the increase in satellite fleet size, what procedures will be put in place? There is still plenty to discover and learn about this area. Satellite companies have promised to link people who are presently unconnected, but the beneficiaries of this connection will be who, exactly? This work has discussed the important quantum technologies, including quantum communication, quantum computation, quantum simulation, quantum metrology, sensing and imaging, quantum control and quantum software and theory aspects. This work can be extended to include important research directions in discussed areas. Furthermore, the comparative analysis of quantum technologies and recent developments can be explored in detail. The proposed work can be extended with satellite and drone-based internet connectivity. Quantum computations can ensure fast and secure communication [157–159]. While quantum computing has the potential to enable new forms of processing, it has also been

proved to pose a threat to the security of the overwhelming majority of cyber-operations in the past. This is another paradigm that is becoming more prevalent in traditional IoT systems. Thus, quantum satellite and quantum drone-integrated internet connectivity will be more relevant for futuristic IoT-based applications like quantum internet for homes. If current security procedures are undermined due to quantum computing, new means of protecting against these risks will be required in the future. Thus, quantum internet using quantum satellites and quantum drones will be more appropriate.

Fraire et al. [4] discussed the possibility that satellite constellations may serve as circular gateways for the Internet of Objects, allowing for creating a global network of DTS-IoT. When constructing the sparse satellite constellation, DTS-IoT takes advantage of the capabilities of LoRa devices to withstand the maximum degree of clockdrift. While the in-orbit infrastructure has been much reduced, latency continues to be a problem in the IoTs networks with limited resources, although the infrastructure has been significantly reduced. This study showed that it is possible to offer global LoRa-based DTS-IoT with just nine satellites in the proper topology if the satellite constellation is configured correctly. With the standard LoRaWAN protocol modes, it would be compatible with a correctly constructed sparse constellation, demonstrating that space-terrestrial IoT integration may be accomplished with fewer resources than was previously analyzed to be possible. In this work, the algorithms are utilized to increase or decrease the size of the satellite fleet until it can fill a maximum coverage gap of 120 minutes in a given time period. This work has conducted performance analysis using coverage function and complexity analysis using the complexity-progressive gradient descent algorithm. This work can be extended to include other algorithms and parameters for analysis. Also, IoT network-related parameters can be considered for performance measurements. These parameters include data transfer rate, delays, goodput and error probabilities. In resourceful or resource-constraint IoT networks, security is always a major concern. Using quantum computers as a test subject revealed that the authentication mechanisms can be immune to the impacts of quantum computers after rigorous testing [158]. When this strategy is combined with a long-term evolution architecture (e.g. quantum satellites or quantum drones), the ability to apply secure and fast communication becomes successful without modifying the underlying base system. Several simulations and performance studies have shown that current systems have much-enhanced security and key management capabilities than older systems. However, quantum computations-based security challenges are always a concern [158,159]. Therefore, post-quantum approaches or efficient quantum computation-based satellites or drones would be more relevant.

Wang et al. [5] discussed that IoT applications benefit from the broad coverage and ease of deployment provided by satellite communication systems, making them an attractive option. In real-time, a dynamic optimum scheduling system for LEO satellites may be created by combining SA-MC algorithms with a new transmission scheduling algorithmic approach. In addition, simulation findings indicate that the SA-MC method suggested reducing the cost value while achieving rapid convergence at the same time. This work has analyzed the performance of a constellation of satellites using cost, throughout, and algorithm convergence parameters. This work can be extended to optimize the algorithm's performance by applying multi-constraint optimization approaches. Further, the concept of the constellation of satellites can be mapped over quantum satellites, quantum drones or hybrid network-based constellations. Dolgopyatova et al. [6] investigated the use of Internet broadcasting for the distribution of media and mass communication information. This work discussed how Internet mass communication security techniques will evolve in the future and how they will stay essentially in the future. This work has ad-

ressed the importance of these security techniques in real-time applications, like how artificial intelligence and neural networks may be used to fight the spread of fake news in the media environment. This survey work can be extended to include a comparative analysis of security techniques and usage of satellites in real-time applications. The security techniques for internet mass communication can be enhanced in the future by using quantum technology like as discussed by Rahman et al. in [160]. A hybrid IoT security architecture is presented, with an additional layer that assures a quantum state. By preserving its status, this condition avoids any malicious acts from unauthorized parties in the communications platform and on the cyber side. Quantum satellites and quantum drones have the power to enable quantum-based communication, which is more secure, faster and covers long distances. Thus, integrating quantum drones and satellites with IoT is the futuristic need for work.

Wang et al. [7] discussed the importance of satellites for the development of smart cities. Here, the primary focus is an infrastructure for allowing large machine-type communications among widely distributed sensors servicing huge metropolitan areas. It is expected that similar infrastructure with the help of satellites will be developed from the Smart City IoTs during the next few years. While the terrestrial IoTs are key components of the smart cities ecosystem, this article looks at non-terrestrial Internet technologies for Smart Cities in addition to the terrestrial IoTs. Following that, the main satellite IoTs technologies and unmanned aerial vehicle IoTs technologies are assessed in independent assessments. The satellite IoTs study is primarily focused on route planning for unmanned aerial vehicles, with both physical and non-physical layer technologies being investigated as part of the effort. This work has not discussed the quantum computing aspects. A distributed computing environment using quantum computers can provide high performance and security. Thus, the design and implementation aspects of quantum computing with intelligent cities can be explored to analyze the possibilities of futuristic smart networks. The distributed computing environment supported by quantum computers can be used in smart cities development in the near future. This work is discussed by Caleffi et al. [161], in which the authors explored how quantum computers may be connected through a quantum internet to achieve exponentially computing speedups. They also outline important future research difficulties and unresolved concerns in the design and implementation of quantum internets. Quantum satellites and quantum drones are other alternatives for providing fast and secure quantum internet.

Among other approaches [8–20], various satellite-based solutions have been proposed for real-time applications. For example, Giuliani et al. [8] discussed the Internet infrastructure using a low-earth-orbit constellation of satellites. In this work, a cost-performance trade-off is performed for Internet routing. This work has examined the properties required for inter-domain routing. In optimal solution-based path aware network architecture design, routing optimization and proposal findings, during theoretical analysis and comparative study, cost of deployment, architecture stability, and latency have been identified as important properties to consider while deploying the Internet infrastructure using satellites. There are various challenges in the content delivery network-based approach, including the highly dynamic nature of satellites placed across the region, fluctuating connectivity due to unstable connections, bandwidth-latency trade-off in data forwarding and high cost and lower bandwidth availability make data traffic difficult to manage. Huang et al. [9] discussed the recent developments in commercial satellite systems. Here, a few commercial satellite communication systems and their properties are discussed. Further, this work also considers the performance parameters into its centre for futuristic usage in real-time applications. This work is a short description of satellite communication systems and can

be extended to analyze and compare satellite features in detail. Zhen et al. [10] discussed the Internet of Remote Things concept with low earth-orbit satellite-assisted 6G. This work has concentrated on ensuring massive random access requests of IoT devices. Initiatives are taken to reduce the overheads and detection mechanisms. Statistical analysis of results provides robust mechanisms and improved systems with better accuracy, success probability, and energy consumption.

Pfandzelter et al. [11] use 6G for satellite and UAV-based wide-area IoT creation. The proposed approach is a multi-domain resource allocation algorithmic approach for improving network efficiency. The simulation of the proposed algorithm confirms its validity and promises its efficiency for wide-area IoT. In [12–15], satellite or constellation of satellites for IoT network development and performance measurements are discussed. These approaches use various statistical methods to improve system efficiency. Further, the interconnectivity of satellites and IoT devices is proposed using fast network connectivity. In [16,17], NOMA and satellite-based IoT concept are introduced. These studies have conducted simulation-based performance analysis. Further, statistical analysis is performed in system design and validation. The analysis shows that the satellite-based IoT approach outperforms compared to traditional IoT approaches in terms of throughput, jitter, delay, and

other QoS parameters for both resourceful and resource-constraint environments. The significant shortcomings in existing approaches [8–17] include (i) no discussions over the usage of quantum computing aspects. Quantum computing can protect futuristic communications from various attacks. In [8–17], no analysis is conducted on the usage of quantum computing aspects to improve the performance or security in the network, (ii) quantum satellites or quantum drones-integrated networks are recently developed to improve the Internet services. These services ensure secure data transmission without interference. Thus, quantum satellites and drones aspects need to be explored. Table 2 shows the comparative analysis of recent studies over quantum and software services to satellite-based IoT or constellation of satellites. The multiple satellite-based solutions supported by quantum technology may increase the efficiency of various real-time applications. Out of which, resources allocation for satellite-based quantum communication is a significant challenge, and the same has been discussed by Bacsardi [162]. Numerous free-space investigations and various out-of-lab tests have been conducted to illustrate the viability of wired-based quantum communications infrastructure. The author outlined the criteria for such a network and detailed our findings on a unique, entanglement-based satellite quantum communications system in this research.

Table 2

Comparative analysis of quantum and software services to satellite-based IoT or constellation of satellites.

Author	Year	A	B	C	D	E	F	G	H	I	Major observations	Shortcoming and future directions	Futuristic quantum-based possible extensions for the proposed approach
Acín et al. [24]	2018	✓	✓	✓	✓	✓	✓	✓	✓	×	This work covers quantum communication, quantum computation, quantum simulation, quantum metrology, sensing and imaging, quantum control, and quantum software and theory domains.	This work can be extended to include a comparative analysis of quantum domains, their challenges and future cope. Further, the performance and security aspects in these domains can be taken up for in-depth study.	Some quantum satellite electronics are inoperable due to radiation [152]. Satellite electronics may benefit from nanoelectronics. QCAs may improve performance here.
Chu et al. [1]	2020	×	×	×	×	×	✓	✓	×	✓	This work has proposed two resilient beamforming methods to lower the overall power consumption of the LEO satellite when the operational channel phase is not favourable. This work describes a multibeam LEO satellite network-based IoTs network with a specific purpose.	This work does not discuss quantum communication or computing aspects for IoT networks. Thus, this work and its power consumption-based analysis can be extended for quantum technology-based solutions. Variation in simulation parameters to analyze the drift in results or error probabilities can be computed.	There is a need for support from LEO satellites in getting the global quantum internet up and running. People who build future quantum Internet satellites should think about how to communicate with each other uplink. It can also be used to think of the network as a group of satellites.
Yang et al. [2]	2021	×	×	×	×	×	×	×	✓	✓	This work has investigated the effects of energy consumption and the number of node variations that remain in the wireless sensor network. Examined the architecture of the 5G mobile communication satellite infrastructure, including the Iridium, Orbcomm, NB-IoT, and Long Range Radio satellites.	This work can be extended to test for real-time applications or environments rather than performing and validating through simulation or theoretical analysis. This study can be extended to include the security or attack analysis of the proposed approach.	Secure IoT and 5G satellite communications employ ECC cryptography. It's time to prepare the Internet of Things for the quantum-based security in future.

Table 2 (continued)

Author	Year	A	B	C	D	E	F	G	H	I	Major observations	Shortcoming and future directions	Futuristic quantum-based possible extensions for the proposed approach
Graydon et al. [3]	2020	✓	✓	✓	✓	✓	✓	✓	×	×	This work has conducted a thorough examination of the history of satellite Internet services as well as the current announcements of new satellite Internet service providers.	This work can be extended to include research directions and challenging Internet services using satellites and quantum satellites. Further, a comparative analysis of existing studies to identify the pros and cons can be performed.	Quantum communication through satellites or constellation of quantum internet can be formulated for real-time applications. Further, the possibilities of multipoint entanglement in quantum internet can be explored in detail.
Fraire et al. [4]	2020	×	×	×	×	×	×	×	✓	✓	This work state that it is possible to offer global LoRa-based DTS-IoT with just nine satellites in the proper topology if the satellite constellation is configured correctly.	More techniques and criteria for research may be included in this study if it is extended. When evaluating performance, it's possible to consider aspects related to the IoT network.	Lightweight security aspects or quantum-based security approaches for LoRa-based Sts-IoT can be explored in detail. Configuration of quantum-integrated solutions is important to explore.
Wang et al. [5]	2020	×	×	×	×	×	×	×	✓	✓	In real-time, a dynamic optimum scheduling system for LEO satellites may be created by combining SA-MC algorithms with a new transmission scheduling algorithmic approach. This work has analyzed the performance of a constellation of satellites using cost, throughput, and algorithm convergence parameters.	This work can be extended to optimize the algorithm's performance by applying multi-constraint optimization approaches.	Satellite-based solutions supported by the quantum satellites will help the advanced applications in providing the data in real-time.
Dolgopyatova et al. [6]	2020	×	×	×	×	×	×	×	✓	×	This work has discussed how satellite-based Internet mass communication security techniques will evolve in the future. Further, the importance of artificial intelligence, neural network and satellites is explored.	This survey work can be extended to include a comparative analysis of security approaches. Further, the usage and applications of artificial intelligence and neural network with satellites can be explored in detail.	The internet mass communication assisted by quantum technology will shortly enhance security (especially with quantum satellites and quantum drones).
Wang et al. [7]	2020	×	×	×	×	×	×	×	✓	✓	While the terrestrial IoT is a key component of the Smart Cities ecosystem, this article looks at non-terrestrial Internet technologies for Smart Cities in addition to the terrestrial IoT's.	This work has not focused on quantum computing aspects for improving the performance and security of smart city networks. Thus, quantum computing aspects for satellite-based terrestrial IoT networks can be explored in the future.	Quantum technology can implement smart city development supported by the use of distributed computing environment. For example, the hybrid architecture proposed (Fig. 11, Fig. 12 and Fig. 13).

A: Quantum Communication, B: Quantum Computation, C: Quantum Simulation, D: Quantum Sensing/Meteorology, E: Quantum Software for Computing, F: Quantum Information Theory, G: Quantum Software for Networks, H: Other satellite and IoT technical aspects, I: Satellites and IoT performance aspects with quantum-related discussions.

4.1. Software characteristics and applications for IoQDs

To understand the interoperability and cross system functionality in IoQDs, it is important to identify the important char-

acteristics of quantum computing platforms, services, or concept designs for IoQDs. Fig. 14 shows the overview of various software applications and assets studies for IoQDs in recent studies. These software applications and assets for IoQDs include quan-

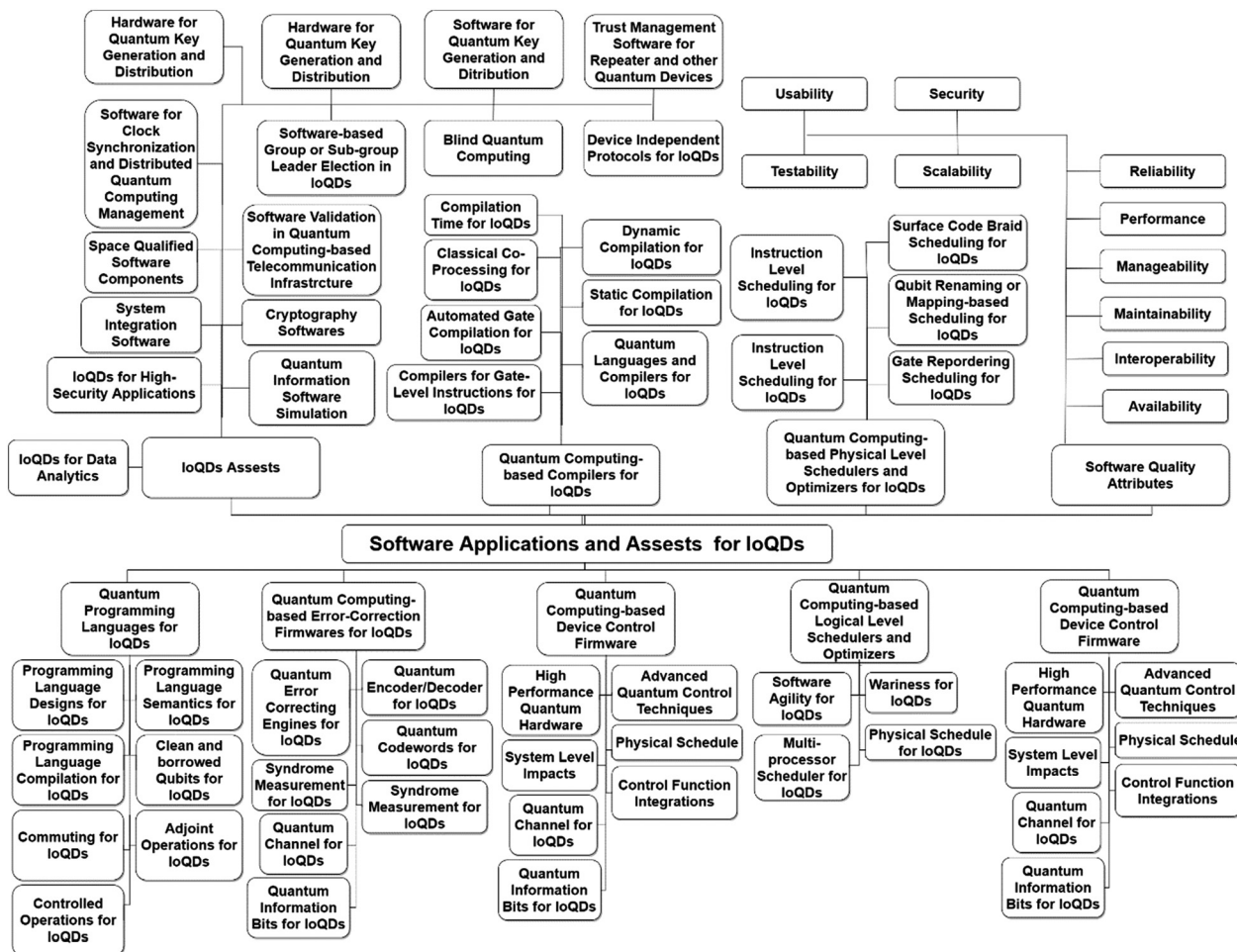


Fig. 14. Important software applications and assets for IoQDs.

tum programming languages for loQDs, and quantum computing-based error-correction firmware for loQDs. Quantum computing-based device control firmware, quantum computing-based logical level schedulers and optimizers, quantum computing-based compilers for loQDs, and quantum computing-based physical level schedulers and optimizers for loQDs. Other assets include software for loQDs for high-security applications, quantum information software simulations, cryptography software, software validation in quantum computing-based telecommunication infrastructure, space-qualified software components etc. In this case, each application or asset requires specialized software services to operate. For example, quantum programming languages could include design, semantic generation, compilation, adjoining, commuting, controlled operations etc. Likewise, quantum computing-based device control firmware requires high-performance hardware, quantum control techniques, physical scheduler, function integration etc. Fig. 14 presents the other classification and sub-classification of quantum related software applications and assets. Some of these applications, assets, and characteristics are briefly explained [44]. For example, wariness for loQDs in quantum computing-based physical level schedulers and optimizers includes quantum radiation effects in the environment, the feasibility of physical level schedulers and optimizers supporting multi-point entanglement for long-distance and multi-object communication, massive security flaws, the inefficiency of quantum computers or devices to handle combinatorial optimization problems which are NP-hard.

4.1.1. Quantum programming languages for IoQDs

Quantum programming languages are important to the program, configuration, and operation of the hardware systems and sub-systems in space computing infrastructure, including loQDs. Various quantum programming languages, tools, simulators, and SDKs have been developed during the last few decades. A few quantum programming languages include Quantum Pseudocode, QISL, Q language, qGCL, Scaffold, QCL, QMASM, and Silq. Fig. 15 summarizes the important classifications of quantum programming languages and how it is important for loQDs. Quantum programming languages can be classified into quantum imperative programming languages (e.g. LanQ, QCL, Q etc.), functional quantum programming languages (e.g. Quipper, QuaFL, QPL, QFC etc.), (quantum domain-specific programming languages (e.g. controlled operations and transformations, clean and borrowed Qubits etc.), multiparadigm quantum programming languages (e.g. Q#, strawberry etc.), quantum circuit support languages (e.g. QWIRE and QuECT), quantum object-oriented languages (e.g. FJQuantum), structural quantum programming languages (e.g. low-level flowchart language) and dynamic execution-based quantum programming languages (e.g. quantum rotations).

With these programming languages, various other quantum computing aspects are directly associated. For example, quantum verification simulators (e.g. ProjectQ, QX, Liquid etc.) can take the help of these languages to design problems and their solutions. IoQDs can use these languages to program circuits and operate as per requirements. For quantum drones applications, the usages of programming languages are: (i) hardware-based programming

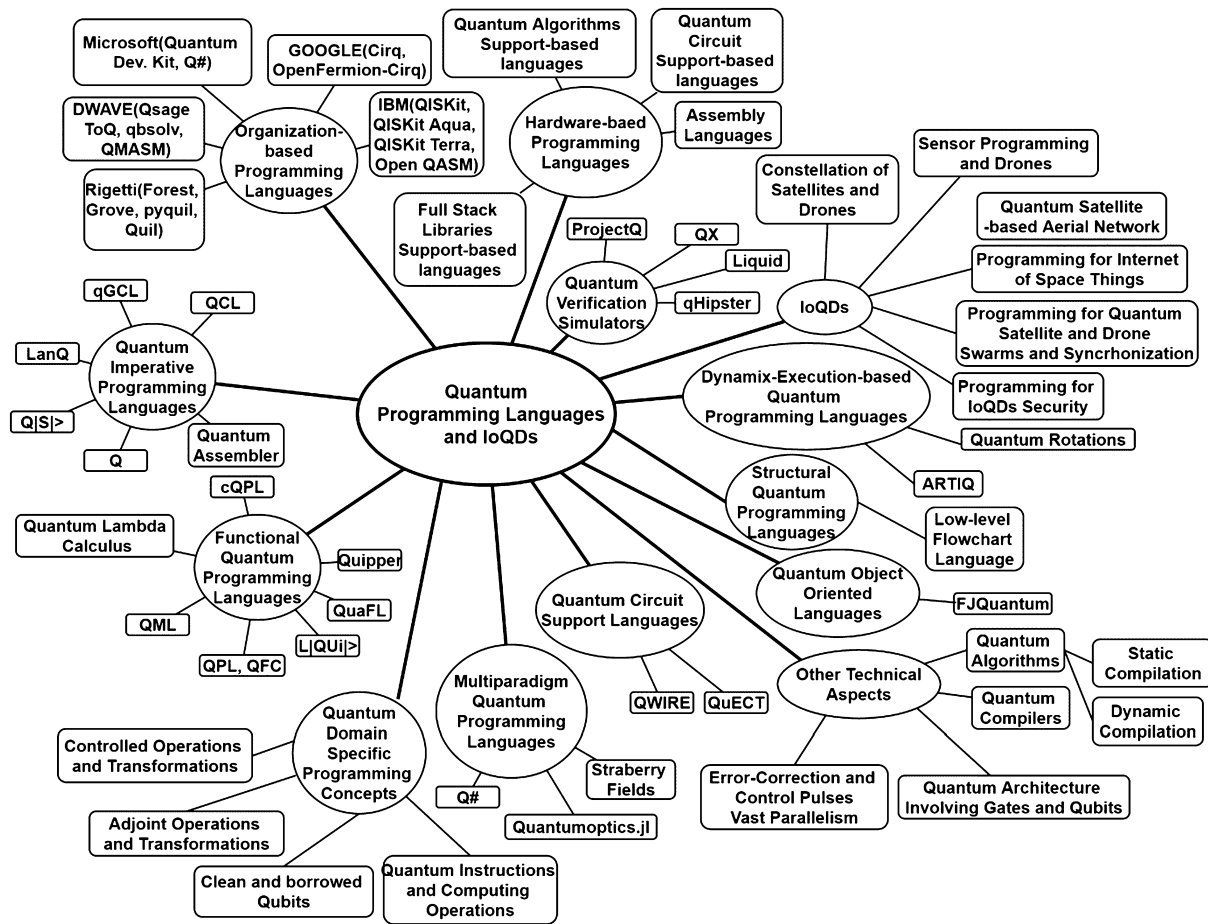


Fig. 15. Quantum programming languages and IoTDs.

languages will be useful for programming quantum circuits, implementing quantum algorithms and writing assembly instructions. (ii) functional quantum programming languages will be useful to operate the drones, (iii) domain-specific languages will be useful for computational, communications and transformation tasks, (iv) object-oriented languages could be helpful in error detection, correction, module-wise function implementation and integrating gates and circuits with different functionalities and (v) imperative languages could be helpful in hardware circuit management and providing necessary classes to drone operations. The important studies conducted over quantum programming languages and their comparative analysis are shown in Table 3.

Pérez-Castillo et al. [45] discussed that quantum computing continues to advance from academic and industry perspectives. This work discussed software modernization in quantum computing and found that there has not been a systematic strategy in tackling this area up until now. According to the study, a method for software modernization is being proposed that aims to reorganize classical systems to operate with quantum systems. This has significant technical and economic ramifications since current system expertise can be reused while new quantum-based initiatives can be implemented faster. This reduces the work required to create new quantum information systems while allowing the knowledge in older systems to be used. Due to this, it is compliant with global regulations. Thus it may be used regardless of the quantum programming language or computer environment being used. This work does not discuss, however, how software modernization would be useful for hardware-based implementation. Programming languages are associated with hardware platforms in

this work. However, devices like drones or quantum drones are missing in scenarios.

As a consequence, software-based approaches for hardware devices need to be focused on in the future. Rojas et al. [46] discuss that the development of many quantum programming languages (QPLs) at the present moment makes it difficult to choose one that has the finest design format for quantum computer developers and users. As quantum computing becomes more widely accessible, researchers will be able to get a deeper grasp of the efficiency of different programming languages. This includes features like developer-friendliness, quantum algorithm optimization, fault-tolerant execution of quantum programs via protocol, and quantum error-correction techniques. This work has surveyed quantum programming languages and their features. Here, a comparative analysis of various studies is performed to identify the assembly language, high-level language, declarative language, and imperative language. Further, quantum programming language design features are identified that include completeness, expressivity, efficiency, and hardware independence. This work can be extended to include quantum language features supporting hardware devices (like drones). Further, software usage and modernization can be explored for the latest advancements in quantum programming languages.

Garhwal et al. [48] discussed the status, a high-level and succinct overview of the current state of quantum programming languages. This study focuses on high-level programming languages having quantum characteristics for quantum computers. This work has listed examples of different quantum programming languages briefly. It has been observed that the majority of quantum programming languages are an extension of high-level programming

Table 3

Comparative analysis of recent studies over quantum programming languages.

Author	Year	A*	B*	C*	D*	E*	F*	G*	H*	I*	Major observations	Future directions
Pérez-Castillo et al. [45]	2021	✓	✓	✓	✓	×	×	×	✓	×	A method for software modernization is being proposed to reorganize classical systems such that they can operate with quantum systems.	In the future, software modernization aspects for hardware-based implementation can be focused on in detail.
Rojas et al. [46]	2019	✓	×	✓	✓	×	×	×	×	×	This work has surveyed quantum programming languages and their features. Further, quantum programming language design features include completeness, expressivity, efficiency and hardware independence.	This work can be extended to include quantum language features supporting hardware devices (like drones).
Garhwal et al. [48]	2019	✓	✓	✓	✓	✓	✓	×	×	×	This article focuses on high-level programming languages having quantum characteristics for quantum computers and comparisons between them and conventional programming languages.	This work can be extended to cover challenges, research directions, frameworks, and the complexity calculation for programming hardware devices.
Heim et al. [51]	2020	✓	×	✓	✓	×	×	✓	×	×	This work has discussed the characteristics, importance, and examples of quantum programming languages. Here, quantum computing aspects are put into the centre to examine the significance of quantum programming languages.	Programming languages and their associations with quantum drones, satellites, and space computing aspects are not yet explored. Thus, programming languages, features, and importance can be analyzed for space computing aspects in the future.
Unruh et al. [52]	2006	×	×	✓	✓	×	×	✓	✓	×	This work has explored the important characteristics, expected features, and examples of quantum programming languages. Here, the focus is drawn toward practical and formal language features and futuristic requirements.	This work can be extended to include the hardware aspects and software solutions for quantum programming languages. Here, parser, assembler, optimizer, and compilers required for quantum programming languages to cover drones, satellites, and other space computing infrastructure can be taken up in the future.
Ying et al. [55]	2010	×	×	✓	✓	×	×	×	✓	×	This work has presented a detailed syntax and semantics for a flowchart programming language, which can be used with high-level quantum programming languages and compiler design.	This work can be extended to include examples of how the proposed flowchart programming language can be used for hardware-based implementations. This may include implementation for space computing-based infrastructure as well.

*Caption: A: Hardware-based Programming Languages, B: Organization-based Programming Languages, C: Quantum Imperative Programming Languages, D: Functional Quantum Programming Languages, E: Multiparadigm Quantum Programming Languages, F: Quantum Object-Oriented Programming Languages, G: Quantum Programming Simulators, H: Full-Stack Libraries Support-based Languages, I: IoTs-related technical aspects.

languages and share similar characteristics. This work can be extended to cover the complexity calculation for programming the hardware devices. The analysis of recent trends can be extended to include futuristic research directions, challenges, limitations, and possible solutions. Heim et al. [51] explored that quantum programming languages are critical tools for turning ideas into quantum computer instructions. These languages are popular because of quantum computers, quantum computing, and their usage in various futuristic real-time applications. Quantum programming

languages are utilized for many different tasks in the quantum computing domain. For example, they are used for controlling current physical devices to calculate the costs of quantum algorithm execution on future devices and even teaching quantum computing ideas to students. This survey discusses an overview of several cutting-edge quantum programming languages, emphasizes their most distinctive characteristics, and provides code examples for each of them in detail. Here, important features of quantum programming language are explored, including commuting operations,

clean and borrowed qubits, adjoint operations, and controlled operations. In the future, loQDs and space computing requirements can be analyzed, and their integration with quantum programming languages can be explored.

Unruh [52] classified programming languages into functional and formal programming languages. Important quantum computing aspects, including requirements of simple and powerful, technology- and platform-independent ability to optimize algorithms and error correction techniques, and emulation (simulation and execution) are discussed. In addition, the challenges and features of quantum programming languages, including quantum branching, continuous classical output measurements, concurrent process execution, infinite datatypes, high-order data types, powerful reasoning, and logic—are discussed. Finally, various quantum programming languages are explored to identify their features, characteristics, and futuristic importance. Ying et al. [55] proposed a low-level flowchart language for quantum programming, particularly for implementing high-level quantum languages. In addition, quantum compilers can be designed using the proposed semantics and syntaxes. Here, examples are taken to discuss the quantum program errors, program and language weaknesses and limitations, and logic errors. This work has presented detailed semantics for flow-chart quantum programming language. This work can be extended to include examples of how the proposed language can be used for compiler design, implementation of other quantum programming languages, implementation pros and cons, and quantum security aspects.

Among other approaches [47,49,50,53,54,56,57], a new quantum programming language - the Qumin language - is proposed [47]. This language's theoretical foundations, lambda calculus, and other practical aspects are covered in detail. Further, algorithms, programming tools, matrices, and quantum fragments of programs are discussed to show their importance. Short surveys over quantum programming languages [49,50], frameworks [53,54], high-level structures [56], compiler design [57], etc., are also discussed to show the importance of quantum programming languages. The quantum programming language area has been widely explored for many real-time applications. However, there is a need to focus on various challenges like (i) designing compilers and assemblers suitable for hardware-based implementations supporting quantum computing, (ii) programming space computing aspects with logical programming, and (iii) securing the hardware infrastructure using logical quantum programming aspects. This may include a constellation of drones and satellites, satellites with ground Internet service networks, long-distance satellite to satellite or drone to drone interconnections, and software modernization for programming aspects and logic.

4.1.2. Quantum computing-based error-correction firmwares for loQDs

Quantum computing-based bit-error correction or QEC mechanisms are widely discussed for QKD, mainly in recent times [38]. For example, Banegas et al. [38] discussed the quantum-resistant security aspects for software updates and proposed post-quantum signature schemes (LMS, Falcon, Dilithium). Here, the focus is drawn toward the implementation of post-quantum cryptographic primitives and protocols for resource-constraint environments. In post-quantum cryptography, a code-based cryptosystem is considered a viable option for resource-constrained settings. However, performance issues in this approach have been identified as an area to address to make it suitable for resource-constraint IoT environments. Svore and Troyer [58] discussed that the QEC mechanism is necessary to extend the lifetime of quantum information. In resource-constraint hardware devices, quantum-based mechanisms are required to have a quantum error bit rate below a certain threshold [58]. Here, quantum computing infrastructure can reduce the error rate quadratically faster. Thus, there is a need to

explore algorithms, approaches, mechanisms, and frameworks that control the error rate for embedded devices. These embedded devices can be integrated into an IoT environment and used in space infrastructure for long-distance communication.

Isaac et al. [59] discussed the use of drones and free-space optical quantum channel for securely exchanging keys over long distances. They examined mobile free-space Quantum secured communication on developing advanced and expansive quantum communication networks. Chen et al. [60] also discussed the importance of free space quantum computing and the need for QKD and QEC. The authors conclude that quantum infrastructure can be reduced with satellites, drones, and vehicles. In this work, an experimentation is performed to implement QKD in free-space communication. Takenaka et al. [61] discussed the case of satellite-to-ground communication using a 50-kilogram microsatellite. In this work, hack-proof quantum communication for QKD is explored using satellites. In experimentation, placing the microsatellite in a low-orbit position to reduce the quantum bit error rate is the goal of this work. Thus, this real-time experimentation demonstrated the feasibility of fast and error-free quantum communication for key distribution. Sun [62] discussed the RFI-QKS approach. Here, the advantages and disadvantages of the proposed system are discussed. In a major observation, it has been observed that source flaws in quantum communications are targeted to be removed. Further, the quantum bit-error rate is reduced. The major challenge of the proposed approach is practical security, which is identified as a significant bottleneck for RFI-QKD.

4.1.3. Quantum computing-based device control firmware

In quantum computing infrastructure, various electronic and device controls are required to operate the embedded machines smoothly. For example, Sebastiano et al. [63] discussed Cryo-CMOS electronic control system for scalable quantum computing. Cryo-CMOS is designed to integrate CMOS fabrication infrastructure. Further, performance aspects are taken into consideration while designing the Cryo-CMOS system. In the study, Cryo-CMOS circuits and systems are considered for complex calculations and facilitating the manufacturing of practical quantum computers. Gebauer et al. [64] discussed the heterogeneous architecture for quantum computing. In this architecture, many embedded devices are interconnected to perform user tasks and data communication. Examples are taken to analyze and explain the working of this architecture. This work has discussed the architecture with timing benchmarks and usage of this architecture from futuristic quantum computing aspects. To operate the architecture, complex control schemes are suggested. Thus, this system integration and operations can be controlled without knowledge of FPGA designs and testing.

Kandala et al. [65] discussed the importance of control software, algorithm, and parameters for experimental optimization of Hamiltonian problems with a hundred Pauli terms in determining the ground state energy for molecules of increasing size. This work has demonstrated the high-performance computing aspects and how different control theories and software aspects help in implementing it over quantum hardware. Von Maurich et al. [66] focused on lightweight code-based cryptography that uses McEliece encryption with QC-MDPC codes for its implementation and error correction. This lightweight approach can perform encryption and decryption in a short time and reduces the key sizes and hardware requirements. This work uses error corrections and a control vector to measure the performance for key sizes and encryption/decryption processes. Among other approaches [67–70], the control function is used differently. For example, it can be used to control parameters for improving learning abilities [67]. Here, learning parameters are controlled to enhance the learning and training abilities and performance in a complex deep learn-

ing network using quantum computers. Further, high-performance quantum hardware, advanced quantum control techniques, system-level impacts, physical schedule, control function integrations, a quantum channel for loQDs, and quantum information bits for loQDs are important aspects discussed in recent studies [68–70].

4.1.4. Quantum computing-based logical level schedulers and optimizers

In the quantum world, quantum computing-based logical level schedulers and optimization processes focus on improving the performance and response time of real-time applications. In literature [71–76,134], various studies are conducted to propose quantum solution-based optimization approaches for IoT environment, performance analysis, energy management, network communications, generating energy statistics and optimizing performance. Table 4 shows the comparative analysis of recent studies over quantum computing-based logical level schedulers and optimizers.

4.2. Software engineering for loQDs

Quantum computing is rapidly evolving, with several practical applications in supply chain management, chemicals, finance and banking sectors, power and agribusiness, healthcare and wellness, and other fields. Organizations will increasingly need to include quantum software in the upcoming decades, and both the academic and practitioner sectors will need to respond to this challenging context. Numerous programming languages and quantum algorithms emerged with the advent of the first quantum computers, yielding impressive outcomes. On the other hand, quantum software has yet to be developed in a systematic and commercial manner. Quantum computing requires the application of software engineering and programming techniques.

The loD [77] is a network architecture that allows drones and people to communicate and command each other through the Internet. In actuality, drones are fast approaching commodity products, permitting any user to fly numerous drones in regulated airspace for various tasks. While technology aids in the mass production of onboard elements such as processors, sensors, memory, and battery capacity for UAVs, the operational limits of these elements hinder and lower expectations. Drones with vehicle and cloud mobility features are available from loD, allowing for remote drone ownership and management.

Researchers [78] are hoping to construct a worldwide quantum Internet based on the transmission of quantum particles to permit ultrasecure communications by encrypting data using secret codes created by the particles. A quantum Internet might potentially enable remote quantum computers to collaborate or conduct experiments that push quantum physics to its boundaries. To transfer the photons, the researchers employed two drones. One drone generated entangled particle couples, delivering one to a ground station while transmitting the other to the other drone. The particle was then sent to a secondary ground station located a kilometre apart from the former. Software-defined communication, which allows for the rapid prototyping of ideas, makes it easy to do this. A more in-depth discussion of the hardware, software, and middleware components of the QC terminal is provided in this work. In software, superdense coding and numerical simulations will be necessary to accurately depict the transmitter and receiver.

The expansion [79] of the QCS industry is being fueled by more significant usage of QCS in the banking and finance infrastructure sector, officials facilitating the growth and availability of innovative solutions, and a large count of cooperative arrangements for technology and development. The software development industry has been good at responding to the need to modify and embrace well-known procedures and technologies for software design and

build new ones that are specifically adapted to this novel programming model. Nevertheless, the creation of an adequate theoretical model for the creation and implementation of quantum programming is essential to the performance. As the first move toward a good quantum software development, this paper [80] outlines the notion of component, which is critical in the software engineering profession and sets essential criteria for assessing the cohesiveness and coupling degrees of a component in the development of quantum programming.

4.3. Quantum software life cycle for loQDs

Considering recent advancements in the creation of increasingly powerful quantum computers, the field of quantum software engineering is gaining traction to provide concepts, principles, and standards for the growth of advanced quantum solutions. Life cycles are used to describe the process of developing, deploying, managing, analyzing, and adjusting software in traditional software engineering. The multidisciplinary nature of quantum computing and similar life cycles give a shared knowledge of creating and managing a program which is crucial. Even though today's modern quantum solutions are almost always hybrid, combining quantum and conventional programming, the quantum application life cycle must include both types of programming. On the other hand, present life cycles are solely concerned with the development of quantum or traditional programs in isolation. Moreover, the different programs must be coordinated, for example, through workflows. As a result, the process life cycle is included in the creation of quantum applications. The authors [81] examined the software artefacts that typically make up a quantum application and described their life cycles in this article.

Since quantum computing is multidisciplinary, an intersection of knowledge of how to build and operate a quantum software program is required. Unfortunately, no current strategy or life cycle encompasses all essential stages that might persist during the growth and implementation process. As a result, in this work [82], the authors provide the ten-phase quantum software life cycle that a gate-based QCS application must go through. In addition, the authors looked at the goals of each step, the methods and instruments that may be used, and any unique challenges or research topics. As a result, the lifespan may be utilized as a starting point for future debates and studies.

The Quantum Software Life Cycle for loQDs is shown in Fig. 16 [171,172]. Step 1 consists of requirement gathering, in which the hardware-related requirements related to loQDs are gathered. Based on the needs captured, splitting of Quantum-Classical is done. Thereafter, the design and architecture of the software are performed. A combination of quantum and conventional components is the outcome of the preceding step. Using these pieces and describing matching subprograms with their capabilities and connections, a structure is conceptualized during the architecture & design process. The design is then fine-tuned using the internals of the various software elements, such as the data structures employed. The resultant specification should give enough information for the individual components to be implemented in the following phase. In designing and architecture, data flow diagrams are created to choose the final design and architecture. The quantum circuitry and classical software artefacts that execute the quantum and conventional problem components that were generated in the previous step are constructed. As a result, a high-level and hardware-independent quantum programming language should indeed be utilized to build the quantum circuits, allowing for subsequent seller hardware procurement. Following the implementation of the quantum hardware independent phase, the quantum application is tested to ensure that it behaves as expected as per the specifications before being delivered to consumers. It involves test-

Table 4

Comparative analysis of studies over quantum computing-based logical level schedulers and optimizers.

Author	Year	A	B	C	D	E	F	G	H	I	J	Major observations	Future directions
Metodi [75]	2005	✓	✗	✗	✗	✗	✗	✗	✗	✓	E	This work has proposed an optimized microarchitecture for scalable computing. The hierarchical and array-based design is used to design a scalable, reliable, and resource-distributed method for large-scale quantum computing.	The proposed design in this work can be extended to incorporate in-depth system-level analysis. Challenges in quantum architecture can be taken up to improve quantum computers' efficiency, reliability, and scalability.
Ajageka [71]	2020	✓	✓	✗	✓	✗	✓	✗	✗	✓	E	This work has developed a model and method to strengthen the quantum computing and overcome the combinatorial complexity issue in solving large-scale discrete-continuous optimization problems..	This work can be extended to include fault diagnosis in optimization problems; e.g., how complex chemical processes can be handled using quantum computing-based techniques.
Kumar [76]	2020	✗	✓	✗	✗	✗	✗	✗	✗	✓	E	This article has prepared a quantum-inspired green communication framework for energy balancing in sensor-enabled IoT systems. Here, the focus is drawn toward evaluating state-of-the-art technologies using quantum solutions.	This work can be extended to experiment with quantum-inspired metaheuristic optimization problems for green communication in sensor-based IoT networks.
Ajagekar et al. [73]	2020	✗	✓	✓	✗	✗	✓	✗	✓	✓	E	This work has proposed a hybrid quantum computing-based partitioning algorithm to find the optimal solution for large-scale job-shop scheduling problems. Here, four classes of optimization problems are considered for analysis.	The classes of optimization problems can be taken up for comparative performance analysis to identify the parameters that improve fuzzy multi-objective optimization problem of oil development.
Ajagekar [74]	2021	✗	✗	✓	✗	✗	✓	✗	✗	✓	E	This work has proposed a quantum computing-based deep learning framework. This framework is tested to identify fault diagnosis for a simulated electrical power system with 30 buses. Here, the applicability, efficiency, and capabilities of the proposed framework are put in focus.	The proposed framework can be utilized for fault diagnosis and performance analysis for other systems as well. In this work, one case study is considered. However, variation in the number of buses could be taken up to analyze the fault diagnosis process variations.
Liu et al. [72]	2021	✗	✗	✗	✗	✓	✓	✗	✗	✓	E	This work has proposed a numerical method of mixed-integer optimal control problems. These problems are taken up to solve the key issues for alkali-surfactant-polymer flooding in oil exploitation.	Optimization algorithms based on quantum computing, including quantum annealing and quantum ant colony algorithm can be extended with the proposed numerical method
Denkena et al. [134]	2021	✗	✓	✓	✓	✓	✓	✓	✓	✗	E	This work has surveyed methods for quantum-computing-based job scheduling. Further, a new flexible job shop scheduling-based algorithm is proposed and evaluated experimentally.	This work can be extended to include practical applications to proposed optimization formulation, objection function with an optimization algorithm, and quantum bit optimization build time.

A: Quantum operation (Quantum Circuit)-based scheduling, B: Job-shop problem, C: Graph-theory for Quantum-based scheduling approaches, D: Flexible Job shop scheduling problem, E: Ant colony-based job scheduling, F: Quantum Simulated annealing-based job scheduling, G: Partial swarm optimization-based job scheduling, H: Tabu search-based job scheduling, I: Other scheduling algorithms, J: Survey (S)/Experimentation(E).

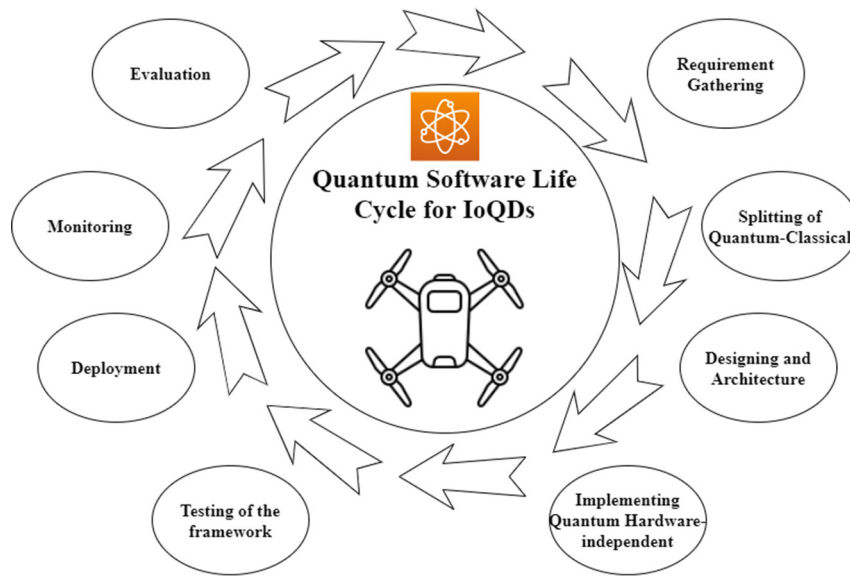


Fig. 16. Quantum software life cycle for IoQDs.

ing all constituent software artefacts, such as quantum programs, classical programs, and workflows, comparable to the execution phase.

During the deployment phase, everything is in place to allow the quantum application to work. Therefore, the performance context for traditional programs is built up. For example, a virtual machine may be established and the appropriate Python runtime deployed on it for a Python code. Quantum applications and procedures must be deployed in the same way. Several of the needed capabilities, on the other hand, may be offered as a service or API, requiring no deployment. The quantum program and its programming environment are then monitored in the following phase. Data is gathered for two motives: monitoring the current state of a functioning quantum program and preserving the data for long-term study, such as to enhance the quantum implementation or to provide provenance, understanding ability, and repeatability. This step necessitates gathering information on the mixed quantum application's software artefacts. The acquired data from the monitoring phase is evaluated in the last phase of the life cycle. The objectives of this stage are to identify any problems that need to be repaired as well as prospective quantum application enhancements. For instance, if quantum algorithms regularly return incorrect outputs, this might be due to a sub-optimal division for today's restricted quantum computers.

4.4. Quantum finite automata for IoQDs

Quantum finite automata provide a solid educational foundation for presenting quantum computation ideas to software engineers due to their relative accessibility. However, early quantum finite automata systems were troublesome in that they did not fully encapsulate the potential of quantum physics, resulting in perplexing outcomes in which a "quantum" computer could not replicate its classical equivalent. Several basic quantum finite automata techniques are presented in this work [83], demonstrating the advantages of quantum computation over conventional processing. Finding quantum frameworks that surpass their conventional equivalents is a key problem in quantum information technology. Automata is a basic computer paradigm with numerous applications in a wide range of areas. It has been demonstrated that when compared to the conventional automata, the quantum variant of the automaton can handle specific problems with a

significantly smaller state vector. A deterministic finite automaton is a bounded machine that accepts or rejects letters as input sequences. The word "deterministic" refers to the fact that a deterministic finite automata executes just one calculation per input string. The authors [84] looked at deterministic finite automata (DFA) and showed that DFA has the same processing capacity as non-deterministic finite automata.

In recent years, simulation activity has risen dramatically from atomic to the macro level and quantum dimensions. Computational chemistry is used to make systems for the functioning and modeling of devices, ranging from atoms and molecules to manufacturing applications. It is affected by the massive growth of computer power and algorithm effectiveness. The modeling of chemical reactions in thermodynamic parameters using conventional automata theory significantly impacted computer engineering. A logical objective [85] is to investigate the chemical processing of information using quantum computing frameworks. The authors [86] offered an improved photonic realization of 1qfa to identify the unary language in the study, which considerably enhances the effectiveness of previous work. The technique uses the polarization level of flexibility of single photons and takes advantage of the ability to identify both lateral and vertical polarization. For lesser amounts of the mean number of transmitted photons, the resultant system surpasses the initial automaton. Furthermore, we have added to the previously discovered results by comprehensively examining the parameters under which such 1qfa can function with great dependability.

4.5. Security and cryptography aspects for IoQDs

The capacity of new technologies like the IoTs to protect data and communications for an extended period is critical to their long-term success. As one of the fastest developing IoT areas [87], IoQDs may provide significant economic and strategic potential for entrepreneurs, carriers, and associated equipment manufacturers. The fast developments in quantum computing, on the other hand, represent a danger to cybersecurity and existing cryptosystems utilized in IoT and IoQDs. Nevertheless, quantum cryptography, one of the most advanced quantum computing applications, is an encryption technology that can offer uncompromising protection even if an adversary has infinite rights and access to a quantum computer.

Massive quantum computers and the increased processing capability they will provide might have devastating cyber security

Table 5

Comparative analysis of technological aspects for quantum satellites and loQDs.

Author	Year	A	B	C	D	E	F	Major findings
Mario et al. [89]	2021	✓	✓	✓	✓	✗	✗	This paper describes guidelines for a relay arrangement involving one quantum CubeSat and two quantum drones placed at different locations across the earth. CubeSat produces and disseminates an interlocked couple to both drones situated over the public cloud, each drone devolves through the clouds with its corresponding interlocked photon, and each drone creates a new interlocked photon couple conserving the interlocked pair produced by the CubeSat.
Hua-Ying Liu et al. [90]	2020	✗	✓	✓	✓	✓	✗	The very first mobile quantum telecommunication via coherence dispersion is addressed in this work. The S-parameter surpasses 2.49 0.06 for all-day multi-weather entanglement dispersion during the day, clear nights, and rainy nights. The system has proven to be resilient against the sunlight for all-day multi-weather entanglement dispersion during the day, clear nights, and rainy nights.
Adarsh Kumar et al. [91]	2021	✓	✗	✓	✓	✓	✗	This study also included the creation of a classification of quantum-related domains based on the rationale of their understanding and a study of each of these domains. These fields' problems and suggestions for further studies are also examined. Finally, this paper provides an overview of a current understanding of a variety of potential innovations that have recently been discovered to be crucial for quantum drones and systems.
Lei He et al. [92]	2019	✗	✗	✓	✗	✓	✗	The authors presented a UAV network design based on mobile edge computing in this study, which helps ensure minimal congestion in the UAV network. Presented a specified verifier proxy blind signature method for UAV networks and demonstrated that it is inherently immutable in the random oracle model under an evolutionary selected message assault.
Abhishek Sharma et al. [93]	2020	✗	✗	✓	✗	✓	✗	One of the main focuses of this study is the complex interaction between UAV enhanced cellular transmission and the IoT. Lessons learned, observations, problems, unresolved issues, and possible trends in UAV communications are all included in the study.
Balwinder Sodhi and Ritu Kapur [94]	2021	✓	✗	✗	✗	✗	✓	The goal of this study was to provide a generic framework for quantum computing systems. Introducing a programming interface for quantum software development. Identifying architecturally important features of quantum computing systems and the influence these features have on different performance criteria and the software development process.
Nivedita Dey et al. [95]	2020	✓	✗	✗	✗	✗	✓	This paper outlines a quantum development framework that specifies the distinct features and capabilities of the quantum development plan, quantum acceptance criteria, quantum systems engineering, quantum software coding and deployment, quantum testing, and quantum software quality strategic planning.

Caption: A: Quantum Computing B: Quantum Communication C: Quantum Drones D: Quantum Satellites E: Unmanned Aerial Vehicle F: Software Engineering in loQDs.

implications. Critical issues such as multiplication and the discrete log, the ostensibly complex nature of which ensures the protection of many widely used frameworks, are well understood. Quantum cyber security is a discipline that investigates all elements of quantum technology's impact on the privacy and security of transmissions and operations. When regarded as a resource for enemies, quantum technology can have a detrimental impact on cyber security, but it can also have a beneficial effect when genuine actors employ these techniques to their benefit. The study [88] may be split into three groups based on who has accessibility to quantum technologies and how far these techniques have progressed. Researchers assure that presently achievable jobs are safe in the first class, while researchers study the novel opportunities that quantum technologies provide in the other two categories. Researchers offer genuine parties access to QT to obtain increased characteristics in the latter category, but we confine this accessibility to currently existing quantum technology. The third scenario considers

the privacy and security of mechanisms made feasible by quantum systems in the long term.

Table 5 shows the comparative analysis of recent studies over other software and hardware technological aspects for quantum satellites and loQDs.

5. Discussion: futuristic scientific challenges for loQD

The computational capacity will keep expanding at a double-exponential rate, according to recent forecasts [100]. This implies that the conventional charge of simulating a quantum circuit will grow exponentially with computational hardware advances, and data volume will probably meet Moore's law, where computational volume duplicates every two years [100]. Consequently, various research challenges and directions for loQDs from a futuristic perspective are discussed in the following subsections.

5.1. Challenges to QKD

This sub-section explains the challenges to QKD, which is the backbone of any quantum cryptosystem. Thus, the focus is on making QKD cheaper and more practical compared to existing implementations [117]. Additionally, increasing its range of communication and data transmission rate are independent variables to be considered in this design. In [106], implementation for a new distance of 421 km using fibre-optics is observed compared to the previous record of 404 km. Improvement in data transmission rate will be an important enhancement to real-time applications. Although the new record provides longer-distance communication, the previous record also ensures a higher degree of practical security. Therefore, efforts are required to improve the security levels. In this context, message integrity primitives and protocols can play an important role. Message integrity to avoid signal interference can be useful for a secure communication system. Further, the integration of lightweight cryptographic primitives and protocols will be useful to enhance the security without impacting the performance much. A comparative analysis of lightweight cryptographic primitives and protocols will give more realistic solutions for real-time applications.

According to the literature, other research challenges to be overcome in QKD-enabled microgrids include (i) improving the resilience against side-channel and denial-of-service attacks, (ii) realistic experiments with QKD systems in real-time; (iii) realistic experiments using semi-QKD or semiquantum microgrids (where only one communication element possesses the ability to prepare, send, receive, and measure qubits); (iv) the development of strategies to point-to-multipoint QKD for microgrids; (v) the development of quantum networks to sharing QKD for any couple of nodes in microgrids; and (vi) the development of new QKD cryptography methods and protocols for microgrids [116].

5.2. Challenges to futuristic quantum communication-based architectures

This section explains the main challenges to quantum satellites and the necessary quantum communication-based futuristic architectures. In [107], the authors discuss extending the inter-continental data exchange experimentation with a plan to launch the satellites in lower and higher orbits. Lower-orbit satellites will preferably be used for key exchange, and higher orbits will be used for secure and fast data exchange. In the plan to launch satellites in a higher orbit, there will be a need to construct a constellation of such satellites. This way, a large volume of data can be exchanged securely. However, cost and performance are major issues in building a constellation of such satellites in a higher orbit. Thus, there is a need to explore these directions in the future. Using ultra-cold cesium atoms, it is feasible to do gravity measurements with very high precision. Other real-time applications may benefit from the same kind of trials that were carried out in existing studies [169]. Here, ultra-cold cesium atoms integrated tiny gyroscope may be useful for providing a cold environment to QDs, which will elongate the operational functionalities of QDs and improve their performances.

Despite their advantages over traditional radio-based global positioning systems, quantum positioning systems based on quantum optics also face the following challenges: (i) the creation of a full device architecture that goes beyond desktop simulation; (ii) the development of technologies for maintaining the quantum signal's entanglement state; and (iii) new technologies integrating classical and quantum positioning technologies [118]. With the use of ultra-cold cesium atomic interference, which is only accessible at very low temperatures, gravity measurements may be performed with high precision. When it comes to boosting the accuracy of QDs,

this attribute is advantageous. This technique was employed in the research of a QPS underwater navigation system [169], which included the utilisation of ultra-cold atoms. It is possible that other real-time applications may benefit from the same kind of trials that were carried out here. Further, ultra-cold cesium atoms integrated with full device architecture for QDs will be interesting to evaluate in future.

Quantum satellites, used in quantum drone networks, have a limitation of communicating with fixed locations only. Further, these satellites are susceptible to many space disturbances. For example, satellites use nanoelectronic devices that produce faults when space particles like electrons and protons strike. Thus, there is a need to investigate the possible solutions for better placement with variable location-based communications. Other open areas to be explored in the future generation of Internet of Quantum Satellite Communications include the following: (i) Digital twins for satellite systems can be used to access and incorporate historical data (e.g., sensor or maintenance historical data, etc.) to improve reliability and safety; to forecast performance (e.g., reaction to critical events, the likelihood of mission success, life span, trigger self-repair); and to automate self-repair, (ii) Intersatellite links enable cooperative satellite swarms and clusters, (iii) Hierarchical aerial networks: the issue is to increase coverage and improve the security of communications, (iv) IoTs in space and planetary communications: challenges include developing a new network layer protocol for the space Internet, (vi) Onboard regeneration and flying base stations: using mobile infrastructure to deliver critical services via drones increases the versatility and interoperability of flying base stations embedded in wireless networks, (vii) Increase frequency reuse and dynamic spectrum control for geostationary and non-geostationary satellites, (viii) Automating satellite networks: developing network-slicing algorithms, overcoming graph dynamics in the context of online network-slicing, and expanding the basic protocol between the data and control planes are all obstacles to overcome, (ix) Novel approaches to satellite asset articulation: advanced procedures for optimizing satellite resource utilization while maintaining QoS, (x) QKD for optical satellite communications: QKD is used to generate space-based waveforms and establish stable optical links, (xi) Utilize machine learning to address issues such as adaptive power allocation in satellite-terrestrial hybrid environments, multi-beam satellite output of non-uniform requests, and multi-beam satellite precoding/scheduling to minimize interference [121].

5.3. Challenges related to quantum drones

This section acknowledges and analyzes the critical challenges to quantum drones or drone-based communications.

The first critical challenge is that QDs require an ultracold environment to operate correctly. With an increase in qubits, power consumption increases which in turn increases the temperature. Consequently, there is a need for an ultracold environment to cool down the device and operate. Feng [169] discussed the importance of an integrated small gyroscope that uses a four-pulse Raman laser to monitor a wide range of inertial characteristics, including acceleration and deceleration. This experimentation can be useful for operating ultra-cold cesium atom, which can be used with QDs to improve their performances. For example, Gravity measurements may be made with extreme precision using ultra-cold cesium atomic interference. This property is helpful in improving the accuracy of QDs. Ultra-cold atoms were used to study a QPS underwater navigation system [169]. Similar experimentation can be useful for other real-time applications as well. Ultra-cold atomic and molecular quantum gases in systems of cold atoms or atom chips are an alternative option to ensure an ultracold environment for QDs in future [24]. Ultra-cold trapped ions could also

be explored to ensure the ultracold operating environment for QDs and quantum satellites [24]. In [200], ultracold atoms produced by evaporative cooling are used for experimentation. It has been observed that these atoms reduce the cyclic rate and improve the accuracy of interferometers. Further, these atoms can be used with atomic sensors based on light beamsplitters.

Second, the development of new quantum algorithms for quantum drones is necessary, especially for those applications where it will be useful. The feasibility of quantum drones in various applications in different domains creates a requirement to explore new quantum algorithms and understand the challenges. These algorithms should be domain-specific. For example, quantum drones used to transmit healthcare, natural disaster mitigation, or military data require higher priority than financial or chat applications. For example, Perumal and Nadar [170] discussed the possibility for systems in the healthcare business to be compatible with a wide range of operating systems and platform configurations using quantum communication. Both the length of time it takes to communicate and the danger it poses to security are changed as a result of this quantum-based architectural variation. Quantum keys will be managed in a low-overhead manner, resulting in medical records that are more secure and less hazardous, which will benefit both patients and physicians. In real-time applications, when compared to traditional approaches, the use of a specialised quantum channel to broadcast the produced key reduces the likelihood of leakage and eavesdropping by order of magnitude of more than ten thousand [170].

Third, the present drone-based experimentation is limited to fixed ground-location communication. Further, the communication involves multiple sub-systems (quantum entanglement emitter and key communicators, processing unit, entanglement source, and laser communicators). This sub-system requires more interfaces between quantum drones and ground-based communication systems to accommodate a large set of applications. Consequently, there is a need to explore the requirements and build the interfaces. These findings deepen our understanding of the challenges to quantum drones or drone-based communications.

5.4. Challenges to security aspects in quantum communications and networks

Potential problems regarding security aspects in quantum communications and networks remain open. In this section, we summarize the main challenges to this topic. First, we observed that quantum-proof cryptography or quantum-safe-cryptosystems must be developed to handle quantum computers or communication-based attacks. These attacks can break or decrypt the encrypted data. Likewise, post-quantum cryptography aspects can be explored [119,120]. Many post-quantum cryptosystems are lightweight and can be integrated with resource-constrained drone devices that are easy to protect from quantum attacks, especially lattice and code-based schemes. In the current state of research, attention is being paid to the development of public-key cryptosystems, which will aid in the resolution of the problems outlined in the current and previous sections of this article. Despite their widespread usage and complexity, these systems are sometimes too costly to implement across a broad range of Internet of Things-related applications and services. As predicted, the shift from secure quantum systems to post-quantum secure systems has already started in preparation for the introduction of quantum computers, which is expected to occur in the not-too-distant future, according to current projections [119]. Designing lightweight post-quantum cryptographic solutions may be more useful and cost-effective for futuristic quantum-based applications. A significant step forward in the construction of a future quantum Internet and the realisation of the potential of such an Internet, according

to researchers, has been made by linking three quantum devices together in a network [120]. In such networks, the security of multi-point connectivity is a major challenge. This needs to be addressed with security primitives and protocols in the future.

Another key challenge is regarding the rates of errors in quantum communications. In qubits transmission, the chances of errors are significantly high. Therefore, there is a need to identify self-correcting codes that can minimize the error and improve the performance of drone-based quantum computations. The performance of error-correcting codes can be improved by increasing the error threshold value and decreasing the overheads of the number of qubit requirements. New types of gravity wave sensors and high-resolution telescopes, for example, might be made feasible by a quantum internet, which would also provide ultrasecure communication with less overhead [120].

Third, there are a number of problems when applying IoT in drones. These challenges are rooted in security issues and authentication of drones and IoD [123–127]. For example, “Sharing information through drones is a newer avenue of research and could be revolutionary for the future of smart city technology” [123, p. 6414]. While recent studies have examined drone-enabled IoT applications, the primary limitations to be solved in realistic IoD are related to (i) covering and geographic area range [123]; (ii) drone-based architecture for smart cities applying IoT [123]; (iii) drone speed reducing the performance of drone-based architecture [123]; (iv) provenance-aware distributed trust protocol [125]; (v) security issues related to untraceability and anonymity [126]; (vi) safety in user authentication protocols [124,127]; (vii) Fog computing model and fixed fog nodes to dynamically adapt in different contexts are a challenge to IoD [128]. Furthermore, drone-enabled IoT led to the revolutionary potential in monitoring tasks [129], and more experiments on this theme are needed.

5.5. Challenges for quantum knowledge base

Quantum technology arose as a result of the second quantum revolution. It entails manipulating and controlling specific quantum systems (for example, atoms, ions, electrons, photons, molecules, or multiple quasiparticles) in order to achieve the normal quantum or the limit on measurement precision at quantum levels. The field's cross-disciplinary nature results in a high level of complexity in knowledge and capabilities to be developed in the workforce acting in the field [122]. Hence, a recognized constraint to be addressed refers to the training and capabilities in quantum technology. We explain the challenges of developing knowledge and training in QW for various application domains.

The significant issues applying loQDs in the QW context to be solved in the future include, firstly, the preparation of a quantum workforce. Training and education are required to prepare engineers with expertise in quantum technology and quantum information science. The second issue is related to the massive quantity of data generated by the Internet of Quantum Technology and loQDs and the challenges in distribution, storage, processing, and appraisal (e.g., unification interface, communication protocols, post-quantum cryptography standardization process, etc.) [122]. Third, there are several military, ethical, and peace challenges involved, such as the creation of international and national QW policies and strategies, QW monitoring for countries, trade (importation/exportation) rules, cyber capabilities (e.g., nuclear submarines), DNA manipulation, disruptive materials for war, and intrusive artificial intelligence. Table 6 shows the comparative feature analysis of the proposed system with existing latest works. Most of the existing approaches are based on classical drones, whereas the proposed approach uses quantum drones.

To the best of our knowledge, the present research is the first survey study to examine the nascent research in telecommunica-

Table 6
Comparative feature analysis.

Author	Year	Topic	Type of drone	Features	Weaknesses
Schirber [98]	2021	Small-scale quantum drone network for text and video message transfer	Quantum Drone	Discussed the quantum drone-based experimentation performed for data transfer. Further, possibilities to extend the implemented network for intercontinental communication with quantum drones and quantum satellites are discussed.	Lack of discussion about loQDs, the constellation of satellites, and their integrated architecture.
Baldoni et al. [101]	2021	Molecular drone for atomic-scale fabrication is discussed, and a new paradigm is proposed.	Molecular Drone	Atomic-scale fabrication is discussed, and the need to replace scanning probe microscopy with a coordination compound is explored. The coordination compound shows a similar feature to the molecular drone.	The focus is more on molecular drone aspects than on long-distance connectivity or message transfer.
Barbeau [103]	2019	Drone swarm activities and applications are explored.	Classical Drone	This article has discussed the importance of swarms of drones and their data analysis using classical or quantum machine learning approaches.	Quantum drones, quantum satellites, and their integrated architectures are not explored.
Liu et al. [104]	2021	Drone-based free space communication is discussed, and entanglement distribution through classical drones as the mobile node is experimented	Classical Drone	This work proposed the free-space link between mobile nodes with optical relay among them. The work is aimed to provide low diffraction in free-space transmission. Thus, mobile quantum network feasibility and link establishment with better performance are observed.	Drones for large-scale connectivity and applications to the real scenario are not discussed.
Liu et al. [108]	2019	Drone-based communication in multi-location and multi-weather compatibility is tested.	Classical Drone	This work focuses on key technologies for local and wide-area coverage using quantum networks and classical drones with high-altitude communication.	Proposals to integrate quantum satellites and quantum drones in a hybrid architecture are missing.
Catherwood et al. [112]	2020	Introduced the concept of IoT for the recovery of commercial and retail delivery drones.	Classical Drone	In this study, LPWAN is used for creating an IoT network that should be able to locate and recover delivery drones.	loQDs could be explored to self-detect the location of crashed drones.
Isaac et al. [115]	2020	A critical system with drone-based QKD on a fibre-coupled polarization modulator is discussed.	Classical Drone	A system with drone-based quantum communication, including entanglement distribution. Distributed quantum sensing and verification processes are proposed in this work.	Drones with satellite and low to high altitude drone-to-drone connectivity must be addressed.
Yazdinejad et al. [123]	2021	Focused on a system having increased security and reduced latency on the loDs.	Classical Drones	Blockchain technological aspects are integrated with the loDs to ensure high security, system transparency, and improved performance.	High and low altitude drone movements can be considered for performance analysis and long-distance connectivity.
Bera et al. [126]	2021	Proposed security scheme for a drone in battlefield surveillance and protection from various attacks	Classical Drones	An access control-based security protocol is proposed for ensuring the security of data communications during drone-based battlefield surveillance.	Consideration of the loDs and security protocol integration for long-distance connectivity is missing.
Wazid et al. [127]	2018	A coordination system among drones for a wide range of military usage and user authentication and key agreement scheme is proposed.	Classical Drones	An IoT network with drone-to-drone communication having security through user authentication and key agreement approaches is proposed.	Long-distance drone-to-drone connectivity and security concerns are missing.
Zhang and X. Li [129]	2020	Drone-based IoT environment for environment monitoring with remote area accessibility is proposed.	Classical Drones	A system of environmental monitoring with drone-to-ground connectivity and data transfer is proposed. This system has the flexibility to efficiently utilize the ground intelligent terminal system for environmental data analysis.	Security concerns and long-distance communicational aspects are missing.
Current research	2021	Discuss the importance of quantum drones, quantum satellites, quantum networks, and advanced architectures	Quantum Drones	Multiple quantum drones-based communication, loQDs formulation, and the constellation of satellites construction are discussed from a futuristic point of view. IN the current study, novel architectures are proposed integrating quantum networks for futuristic applications.	Implementation of these aspects could be explored in more depth.

tions technology on loQDs, proposing novel architectures that integrate and create quantum networks enabling futuristic implementations. In this research, the focus is drawn on secure long-distance communication using quantum communication and QKD. At the same time, the existing drone-based studies are more focused on developing security systems using cryptography primitives and protocols for various applications. For example, Schirber's [98] approach of constructing a small-scale drone network with enhanced security can be extended with loQDs for real-time applications that need faster and more secure communication. Small-scale networks give provision to private and public organizations to create a private network and add a layer of security. Baldoni et al. [101] idea of the molecular drone may be useful for loQDs network construction with the least overhead and higher security in future. For example, drone-based applications discussed by Barbeau [103] can be tested with molecular drone-based loQDs. This might reduce the overheads, increases security and elongate the network connectivity. The idea of free space communication [104] and multi-point communication [108] can be useful for real-time multi-location

weather information collection [129], battlefield and other military purposes [12,127], and supply chain management [112] with enhanced security [123]. Likewise, multi-point free space communication using loQDs can be helpful for many space-related data collection and applications.

6. Future research directions

The evidence presented in this study suggests a number of key future research directions to pave the research in the area [21–23, 25–43]. Table 7 presents the important future research areas and the related research challenges.

7. Conclusions

Given the emergent quantum age, this study examined the current status of a nascent field of research on drone-related futuristic technologies, namely QD, the loQDs, and CQS. We observed that the loQDs and CQS are recognized for real-time applications. It is

Table 7
Future research agenda.

Topic	Subtopic	Future research directions
Challenges to QKD	Resilience against side-channel and denial-of-service attacks	With Quantum computing-based drones and satellites, can futuristic QKD be protected from interference, side-channel, and denial-of-service attacks? Faster and more secure communication properties of QKD and quantum computations can do this. Thus, this direction can be explored in future.
	Realistic experiments with QKD systems in real-time	Can real-time QKD and Quantum communication experimentation be established in the multi-country scenario? Presently, communications have experimented between two countries. However, experimentation with 3-party communication is performed in short-range communication. Thus, these experimentations can be extended for secure long-range communication as well.
	Realistic experiments using semi-QKD or semiquantum microgrids (where only one communication element possesses the ability to prepare, send, receive, and measure qubits)	As discussed earlier, the most secure long-range communication experimentation is between two continents and two parties. However, this experimentation can be extended for multipoint communication as well. With QD and satellite-based technology and the feasibility of communicating Qubits in multipoint communication, can secure multipoint communication be achieved in the future? How to achieve secure multipoint communication?
	Point-to-multipoint QKD strategies for microgrids	In the past, it has been proposed that quantum information can be transmitted without transferring or destroying the quantum mechanics principle that underlies the transmission of the information. This is a significant development to make the Quantum internet feasible. Can a similar network's design and development be explored in more detail in future?
Challenges to Futuristic Quantum Communication-based Architectures	Quantum network's design and development to share QKD for any couple of nodes in microgrids	Can building out the QKD links (based on entangled photon pairs) and their highly specialized switches be ensured in future? This development can lead to secure Quantum cryptography protocol development.
	The development of new QKD cryptography methods and protocols for microgrids	Can quantum architecture components, including quantum complexity theory, quantum algorithms, quantum algorithm implementation, interconnection technologies and floor planning, Qubit storage and gate technology, be extended in nearby future such that the quantum architecture outperforms desktop simulation? How do we develop these simulations?
	Design and development of a full device quantum architecture that goes beyond desktop simulation;	Storing and maintaining the quantum signal's entanglement state requires a strongly correlated electron state in the phase state of a state variable, molecular, spintronics, ferromagnetic and quantum process technologies development in designing a device, new formats to formalize quantum state in data representation, and reconfigurable quantum state in quantum computer architecture. Is it possible to develop such process terminologies for quantum computers and computing?
	Design and development of technologies for storing and maintaining the quantum signal's entanglement state.	There must be some form of interaction between the user and a real-world quantum computer to reap its benefits. In order to use quantum mechanical systems in a real-world context, they must be carefully managed. The four "layers" make up a functional quantum/classic hybrid computer architecture are briefly explained as follows: (i) The quantum layer is responsible for the implementation of qubits and quantum computing, (ii) The control processor layer, which operates at cryogenic temperatures, is in charge of controlling and measuring the quantum bits, (iii) The qubits in the system are controlled and measured by an insulating layer operating at or near cryogenic temperatures, (iv) The host processor layer of a computer system is generally responsible for handling user interfaces as well as access to large storage arrays and networks. Is it possible to design similar layered architecture in nearby future?
Challenges to Futuristic Quantum Communication-based Architectures	Design and development of hybrid technology integrating classical and quantum positioning.	

(continued on next page)

Table 7 (continued)

Topic	Subtopic	Future research directions
Challenges to Quantum Drones	Tenacity of Quantum Drones	Drone batteries, which are the thickness of bricks, are hefty and rapidly depleted. Furthermore, gasoline engines are loud and generate pollution. Are there any alternative solutions that can power the quantum drones batteries to make it more efficient and less bulky?
	Navigation problems in the absence of GPS signals	When GPS signals are unavailable, the Quantum drones face problems in navigation. Therefore, there must be a different method to determine the location. Is there any technology that can assist the Quantum Drones in the loss of GPS signals?
	Obstacle detection and analysis of traffic	Without any assistance from human eyes, a drone must fend on its own. The most significant sensor on the drone is the radar. It navigates the drone past other aircraft and obstructions, maps the landscape, locates safe landing pads, determines its height, and even recognizes poor weather. However, radar is not that sufficient for the detection of objects. Is there any other technology that can assist radar for traffic analysis?
	Clustering of Drones	In executing complex tasks, cooperative unmanned robots provide efficiency and reliability. But the clustering of drones raises many challenges in their synchronous functioning. Therefore, there is a need to work on the latest technologies that help in the swarming of drones.
Challenges to Security Aspects in Quantum Communications and Networks	QKD networks lack point-to-multipoint methods	The present QKD network's key distribution function only supports point-to-point key exchange protocol and lacks a P2M method. Therefore, this may lead to a security flaw in Quantum communication. Is it possible to design a secure protocol for the point-to-multipoint methods?
	Numerous techniques consume many quantum node capabilities.	To distribute the security session key for limiting the distance for qubit communication, the QKD network uses the hop-by-hop technique. Is it possible to implement a secure technique for session key distribution apart from the hop-by-hop technique?
	There is no appropriate security connection between conventional end-users and quantum processors.	Considering quantum node building costs are still extremely expensive, it is not possible to provide each end user with a specialized quantum node for utilizing the QKD network's services. As a result, one quantum node must be shared by several end consumers. Is it feasible to provide secure, and customized quantum nodes?
Challenges to Quantum Knowledge base	Small scale quantum drone network for text and video message transfer	Small scale quantum drones are required to share internet connectivity with local organizations or users. These drones will be able to ensure faster and more secure multimedia information. Is it possible to design such compact quantum drones capable of carrying quantum devices for secure communication establishment? This needs to be addressed in future.
	New paradigm related to the molecular drones for atomic-scale fabrication	With the feasibility of quantum drones and their usage in secure communication, designing new computing and communication paradigms, and short and long communication. Can a molecular drone be integrated with a quantum drone to perform rapid atomic manipulation and fabrication? This development can bring vast applications, including proof-of-concept devices, verifying logic patterns, storing atom information, etc.
	Drone swarm activities and applications	Multiple quantum drone flying and their connectivity can provide faster communication in local areas, including cities, towns, or regions. Thus, drone swarm-based activities and their application should be investigated in the future.
	Increased security and reduced latency with the IoTs.	Quantum drones were designed to ensure secure and faster communications. IoTs will be able to provide similar features and collect and share environmental information. Can this advantage be explored for real-time applications in nearby future?
	Drone-based IoT environment for environment monitoring	Quantum computing is widely explored in environment monitoring and cleaning. For example, quantum drones with the capability to clear air would be helpful in highly polluted cities. Can the solution be proposed to implement this scenario and analyze the performance in future? This research avenue requires more investigation.

possible to affirm that the scientific research examining the opportunities and risks involving QD, IoTs, and CQS has just started, although it has become an increasingly relevant debate.

Thus, while recognized contexts of applications include agriculture, shipping, delivery, military, search and rescue operations, and the Internet of “molecular drones,” among others, new potential and unconventional possibilities of use include, for example, IoTs for traffic engineering management, climate change demands, and confidential and encrypted communication. Our findings demonstrate that there are a number of critical and open questions of societal and scientific interest that need to be addressed. As a result, future research projects may be conducted from a range of specific perspectives (see Section 5). Findings contribute to the literature and future practice in the field, providing distinct insights

and new potential applications of drone-related futuristic technologies enabled by the Internet.

Concerning the disruption caused by quantum drones, an important aspect is establishing trust among drones for collaborative efforts. In swarms of quantum drones, the probability is always associated with qubits used to exchange information. Using qubits, it is possible to represent the required confidential information with lesser bits, but the probability of successful message exchange in dynamic situations is a significant concern. This work has compared the classical drone features with the quantum drone.

This study proposed and discussed quantum satellite-based architectures that use low and high-orbit drones for long-distance communication, as well as various applications of quantum drones. In the future, quantum satellite and quantum drones-based architecture for other applications should be explored. Further, the inte-

gration of security aspects like blockchain and quantum-safe cryptosystem could be explored in detail. In sum, this study encourages scientists, governments, and organizations toward further practical understanding and theoretical development regarding how futuristic disruptive technologies can be applied to solve societal and strategic problems.

Other implications and contributions derived from this study are briefly explained. First, specific tasks will likely be done much faster on quantum computers than they would be on standard CPUs if quantum computers are used instead of traditional CPUs. In order to achieve success in this endeavour, the creation of a high-fidelity processor capable of conducting quantum algorithms in an ever-expanding computing environment is essential. Small scale quantum computers can be used in space computing devices as well. Second, according to the NASA research, quantum technology in the form of quantum sensors should be used better to understand the interior structure and dynamics of the Earth. According to this notion, quantum interferometry is the mechanism that would be accountable for the fabrication of these sensor devices. When there is interference in the background noise, interferometry is used to extract information from the interference by combining it with other information from the interference. In this case, IQDs would be helpful in giving more accurate results with secure and faster communications. Likewise, other applications can take advantage of quantum computing. Finally, scientists have now used drones to communicate entanglement, a kind of quantum coupling. Photons were transmitted to two locations a kilometre apart by researchers from Nanjing University in China. The proposed single or multi-bit qubit entanglement can be useful in creating networks (like quantum internet). These networks can be private or public to secure communications. The findings of this study may help better understand how potential and advanced applications of QD, IQDs, and CQS can be applied in new contexts in real-time applications in strategic areas of societal interest in the quantum age.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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